

Broadcast Technology UG1

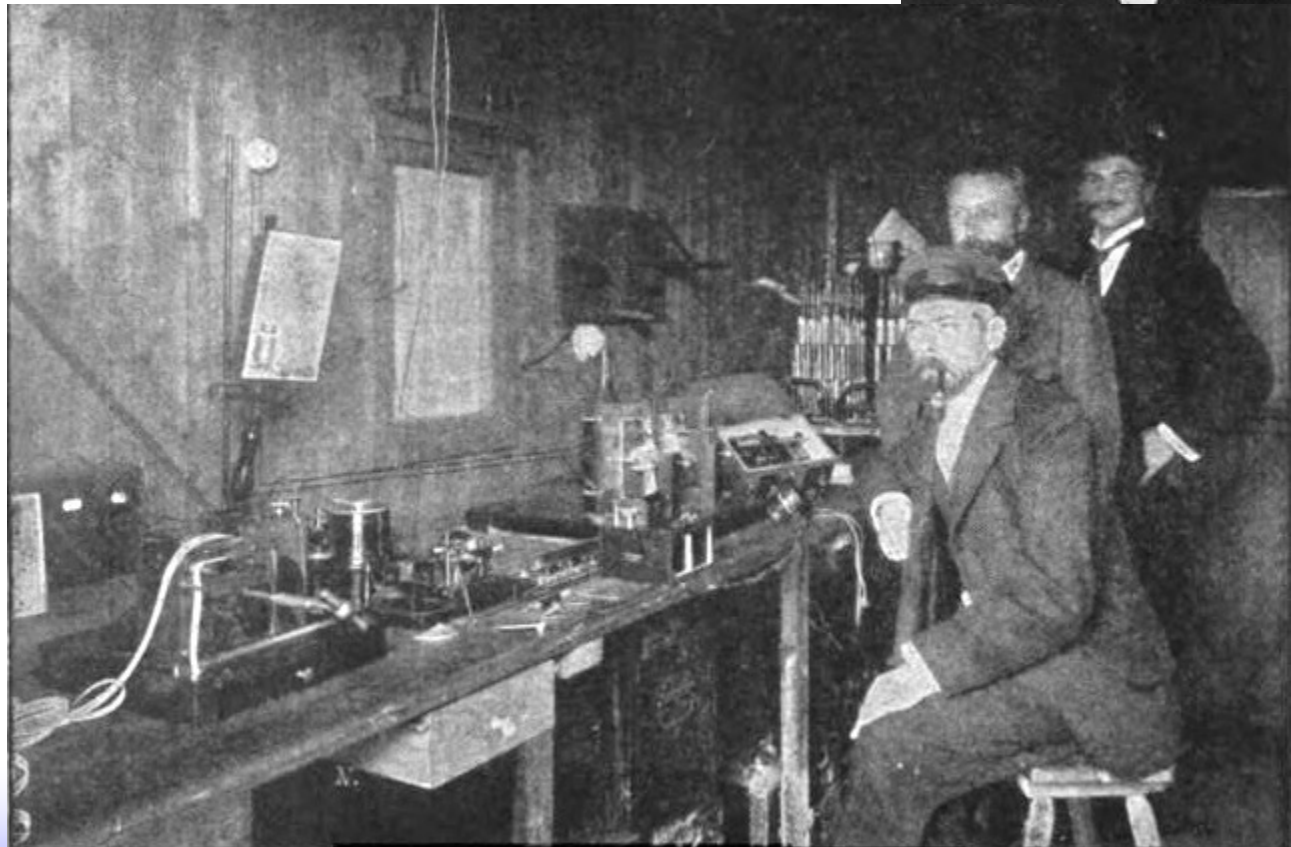
The Legacy of Analogue

History

History – Ferdinand Braun – CRT

Ferdinand Braun:

- 1890 – developed the Cathode Ray Tube (CRT)
- 1897 – developed the Cathode Ray Oscillograph
 - ... the precursor to the radar screen and the television tube
- 1907 – First use of cathode ray tube to produce the rudiments of television images
- 1909 – shared the Nobel Prize for physics in 1909 with Guglielmo Marconi for his contributions to the development of wireless telegraphy



John Logie Baird – Basic TV

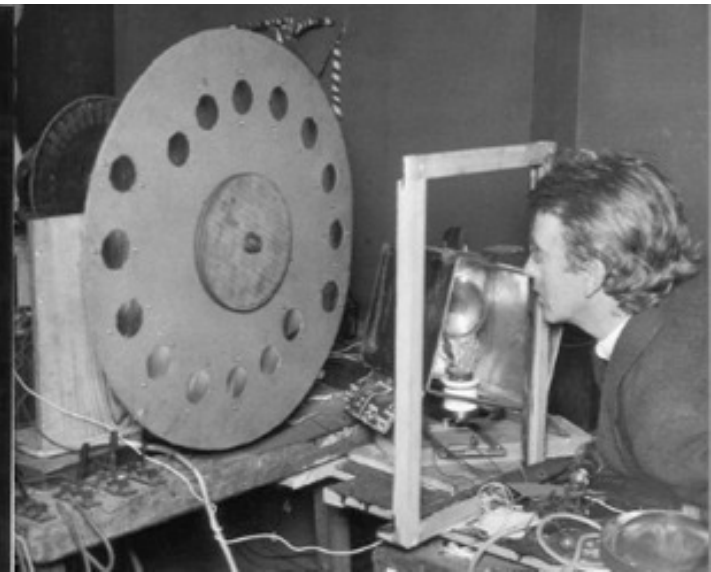
- Oct 1923 – the first person in the world to demonstrate true television, recognisable images, instantaneous movement, correct gradations in light and shade
- Scanning was done mechanically with a Nipkow disc. The first 30 line picture transmitted was a Maltese cross
- 1927 demonstrated video recording
- 1928 transatlantic television
- 1937 broadcast of good definition colour pictures
- 1941 stereoscopic television in colour
- 1944 multi-gun colour television tube



Baird's televisor, 1925: before transmission, the received image at left has been scanned by a spiral of revolving lenses, shown at right with inventor Logie Baird. Photo: Hulton/Getty

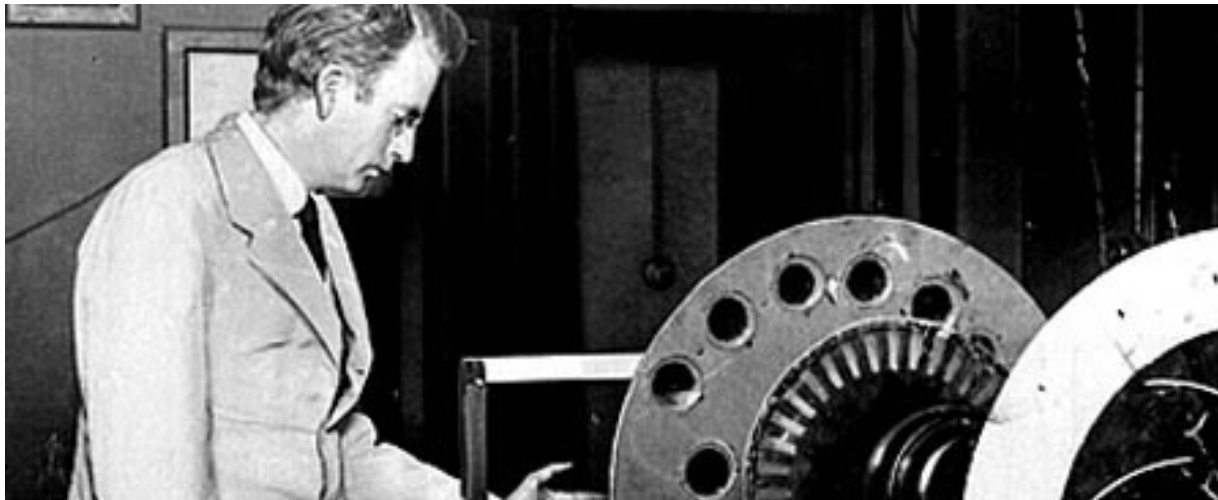
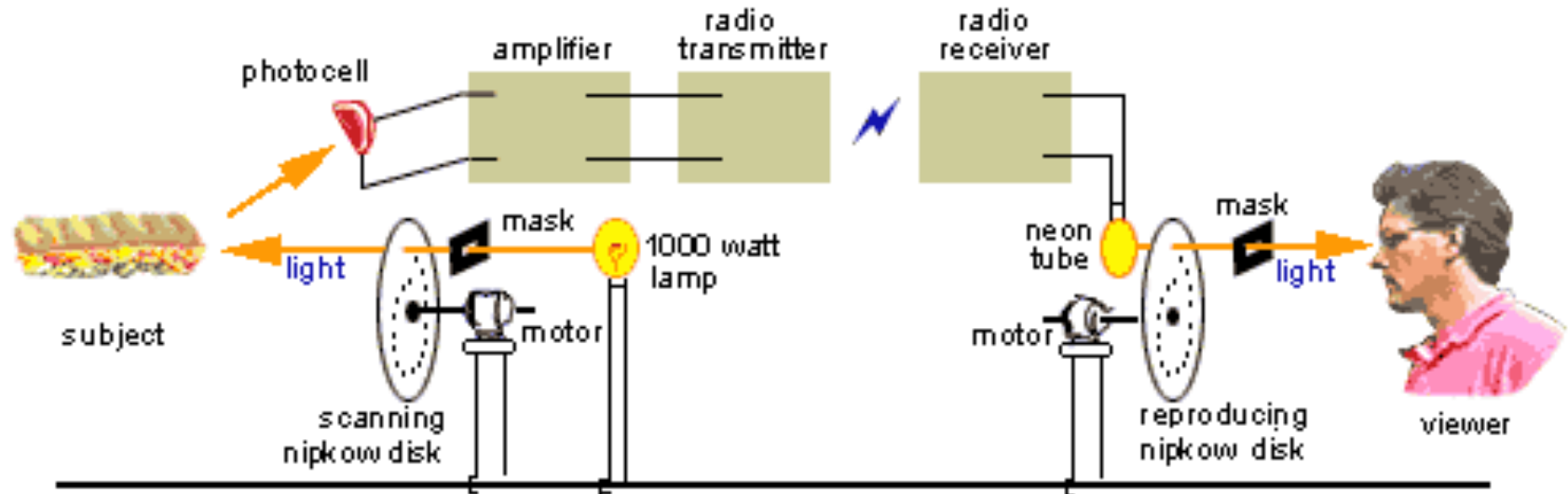


left – early tests were with this ventriloquist's dummy



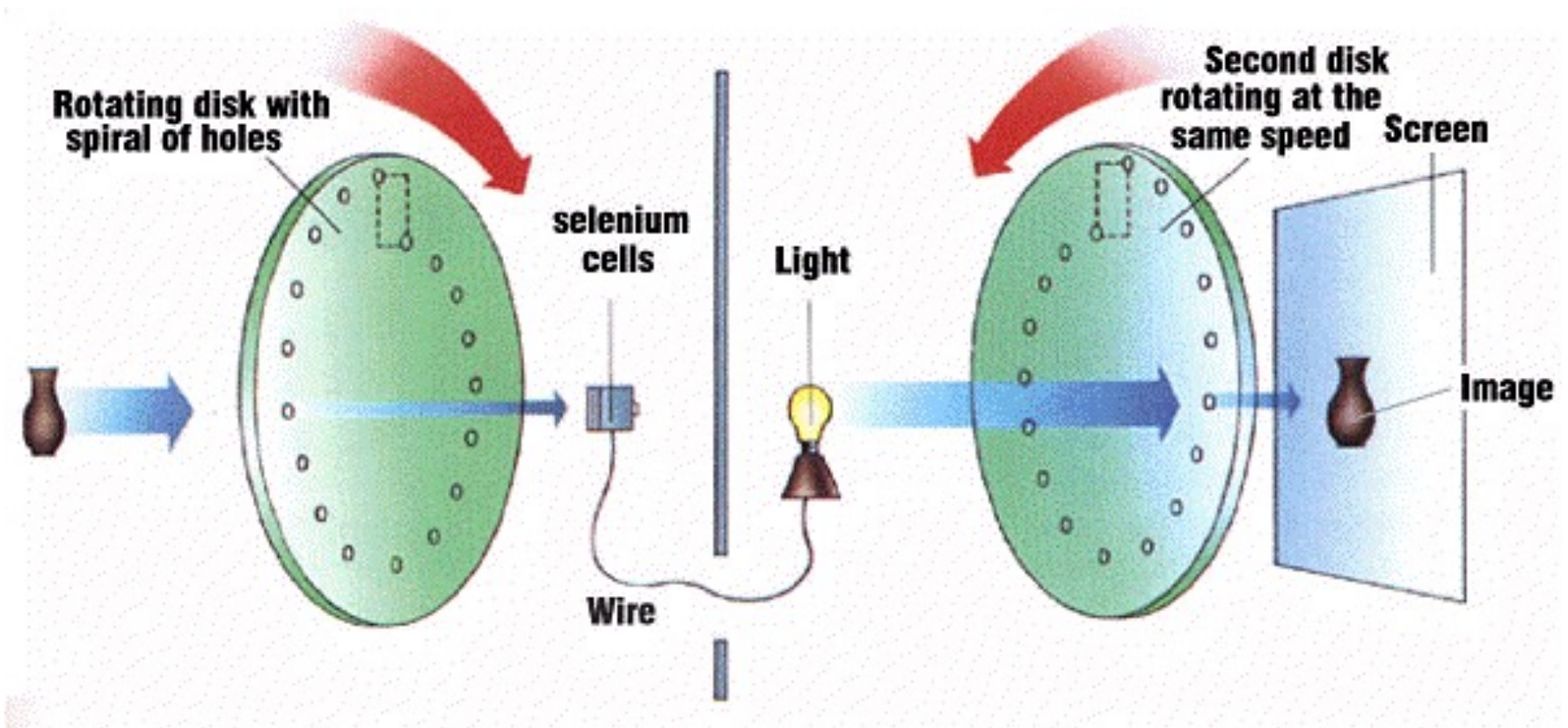
30 Line Mechanical TV

Baird's Mechanical Television System



Early Mechanical Approach to TV

Mechanical Nipkow discs were used to scan the image and reconstitute the image at the receiver. PE cells were used to capture the image. The problem was synchronising the disks.



Electronic Television – Farnsworth

Philo Farnsworth

- 1922 (at Age 14) concepts of Electronic Television
- Demonstration, Sept 1927:
 - painted a square of glass black and scratched a straight line on the centre
 - The slide was dropped between the *Image Dissector* (camera tube Farnsworth invented earlier that year) and a hot, bright, carbon arc lamp.
 - receiver – saw the straight-line image
 - as the slide was turned 90 degrees, image moved
 - the first all-electronic television picture ever transmitted.



Vladimir Zworykin – Iconoscope

Vladimir Zworykin of RCA

- 1923 – patent application for a camera device
- 1933 – developed a camera tube he called an Iconoscope
- Zworykin submitted his patent application first ...

... but after many years of legal battle Farnsworth was acknowledged as the inventor of electronic television.

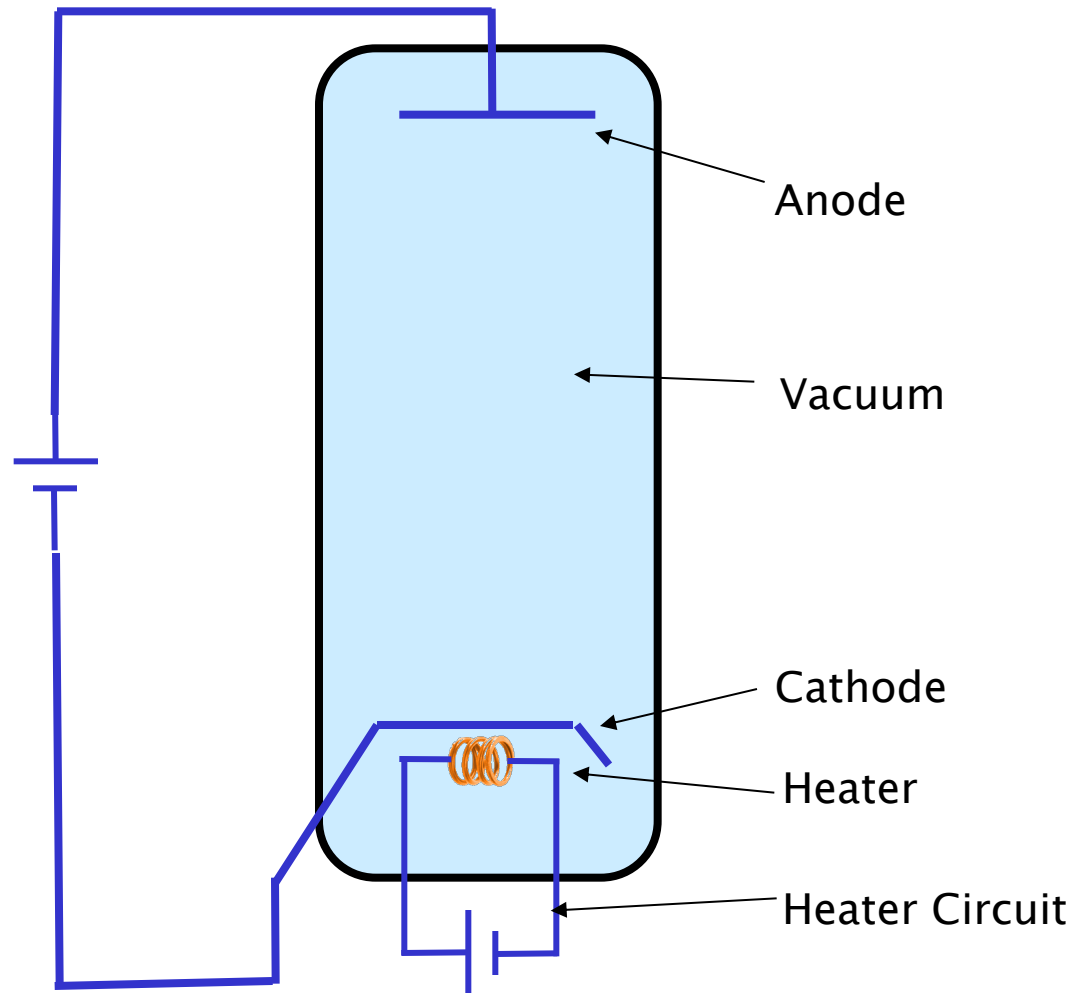
- end 1923 – he produced a picture display tube, the "Kinescope"



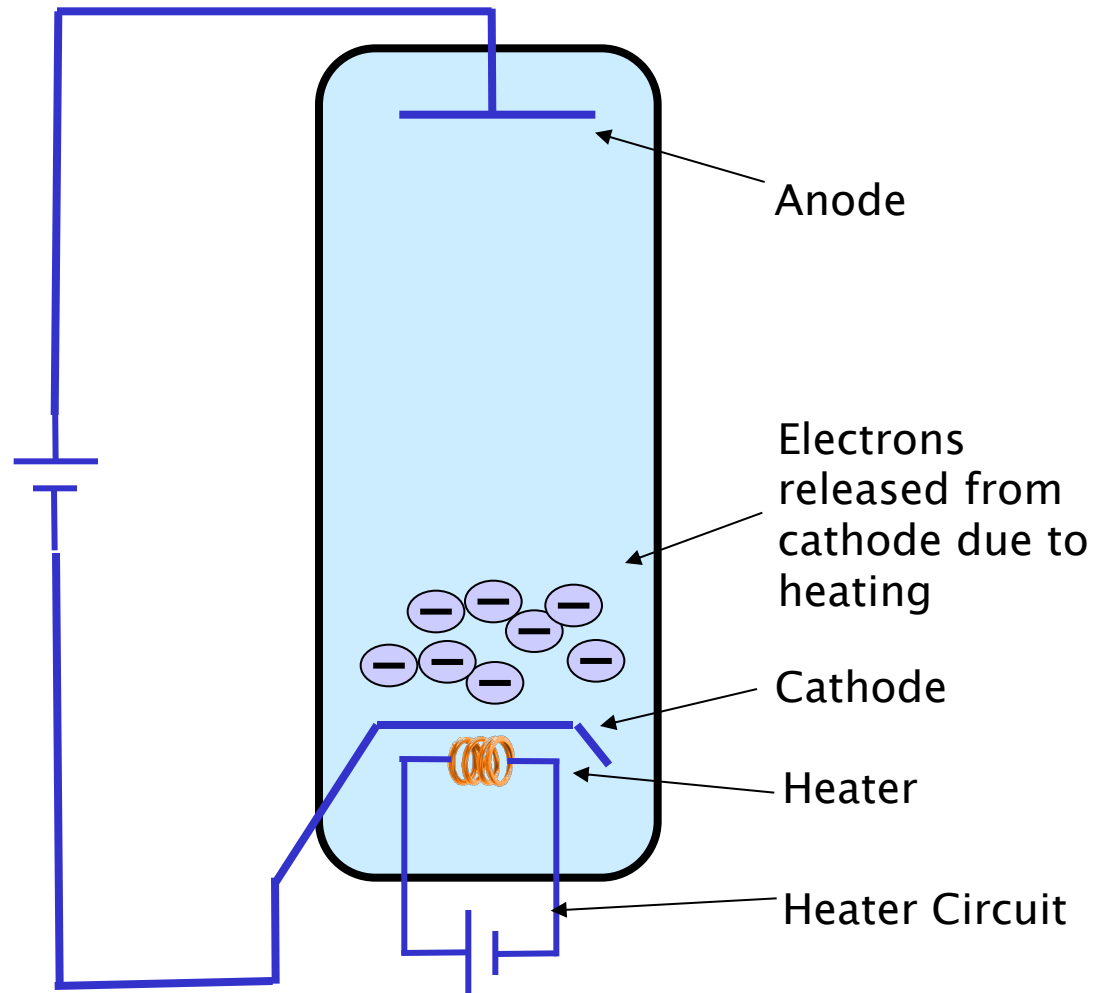
Analogue TV Technology

- mostly history
- but many elements remain in digital systems

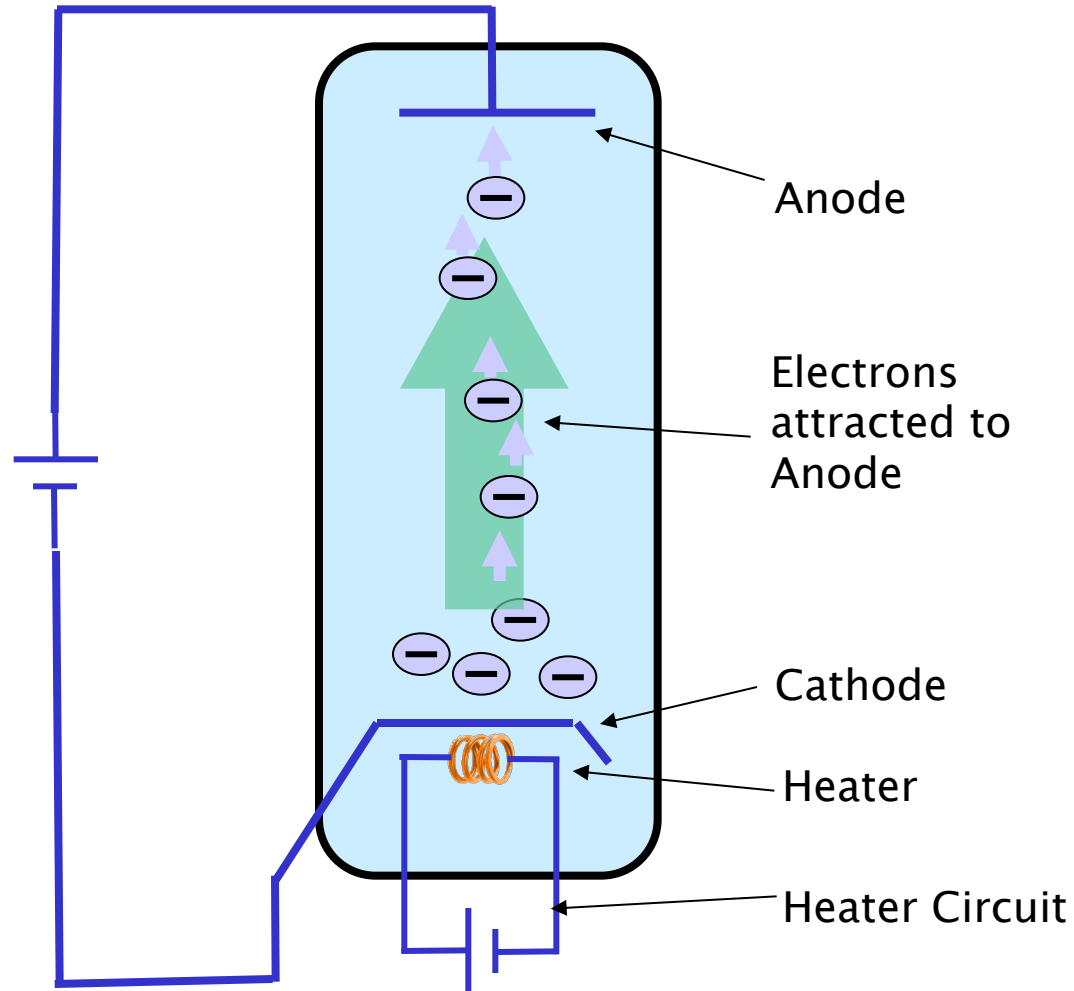
Thermionic Valves



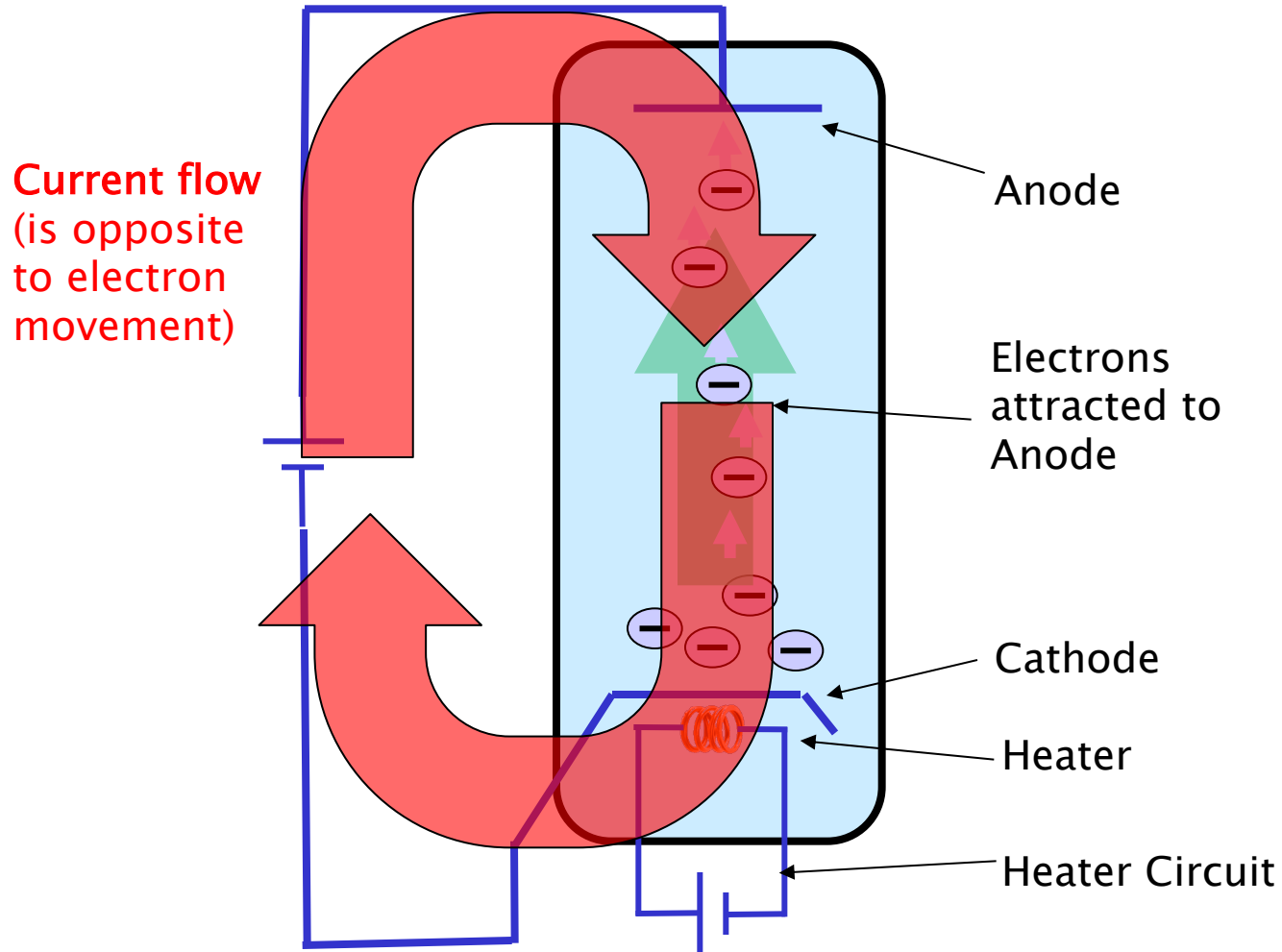
Thermionic Valves



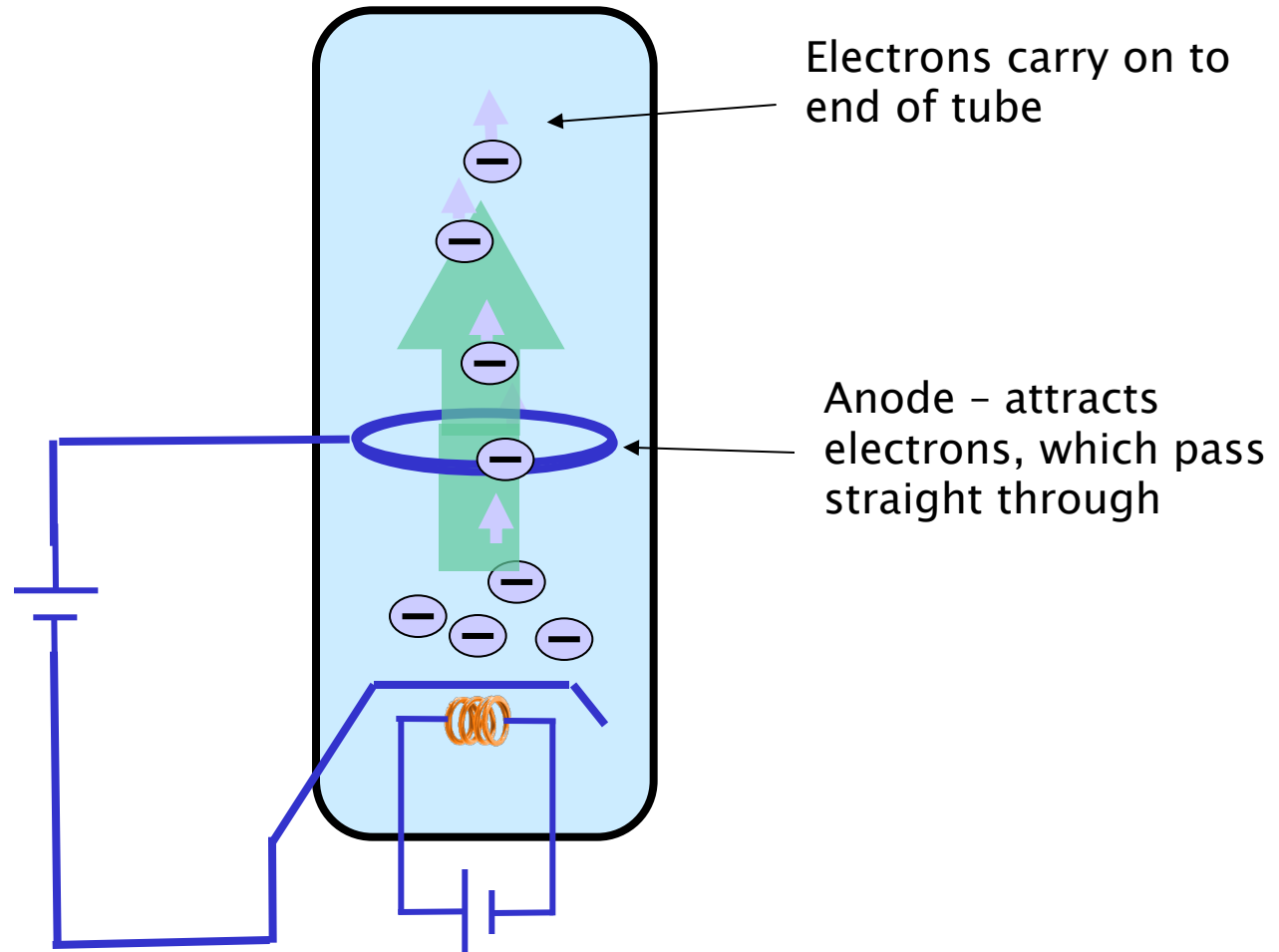
Thermionic Valves



Thermionic Valves

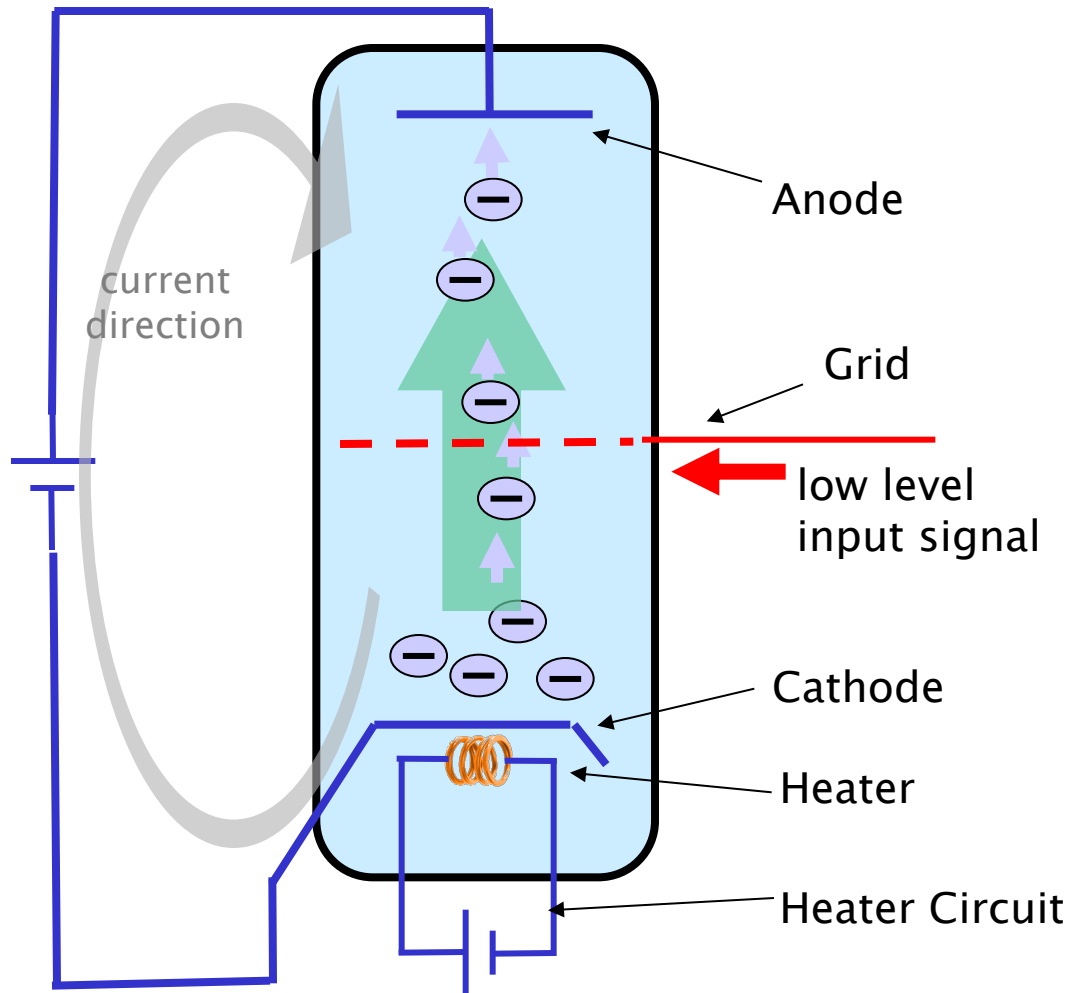


Thermionic Valves – Triode



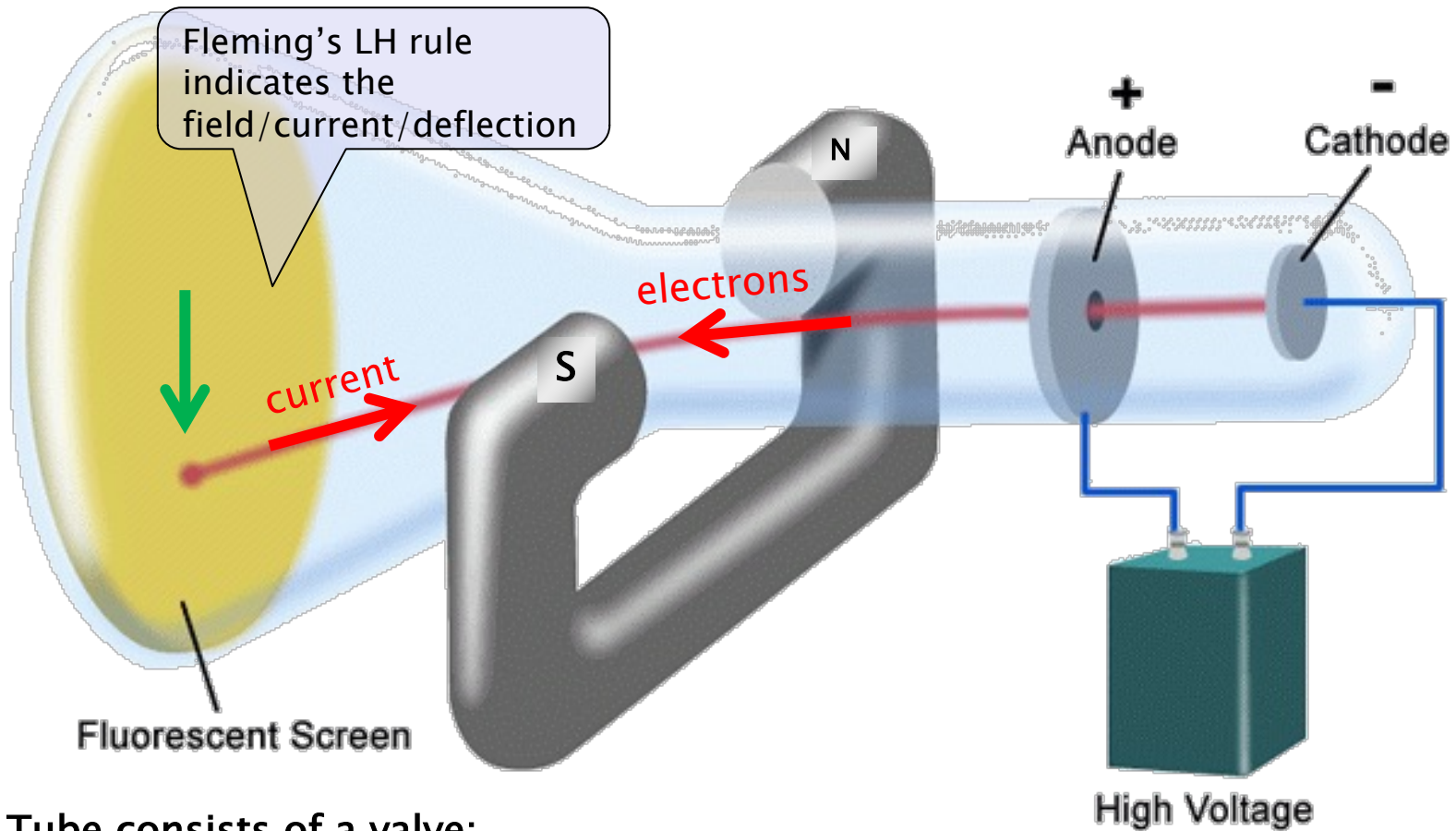
Basic Amplification Circuits

Simple Triode (amplifies)



- A current flows through the valve
- This is initially set up as a “quiescent condition” by circuitry
 - circuit *bias*
- The (large) anode–cathode current can be varied by a small voltage applied to the grid
 - grid voltage varies as the signal input
 - output taken from anode
- i.e. a small varying input voltage will cause much larger variations in the current flowing through the valve
- hence ... amplification

Cathode Ray Tube – CRT

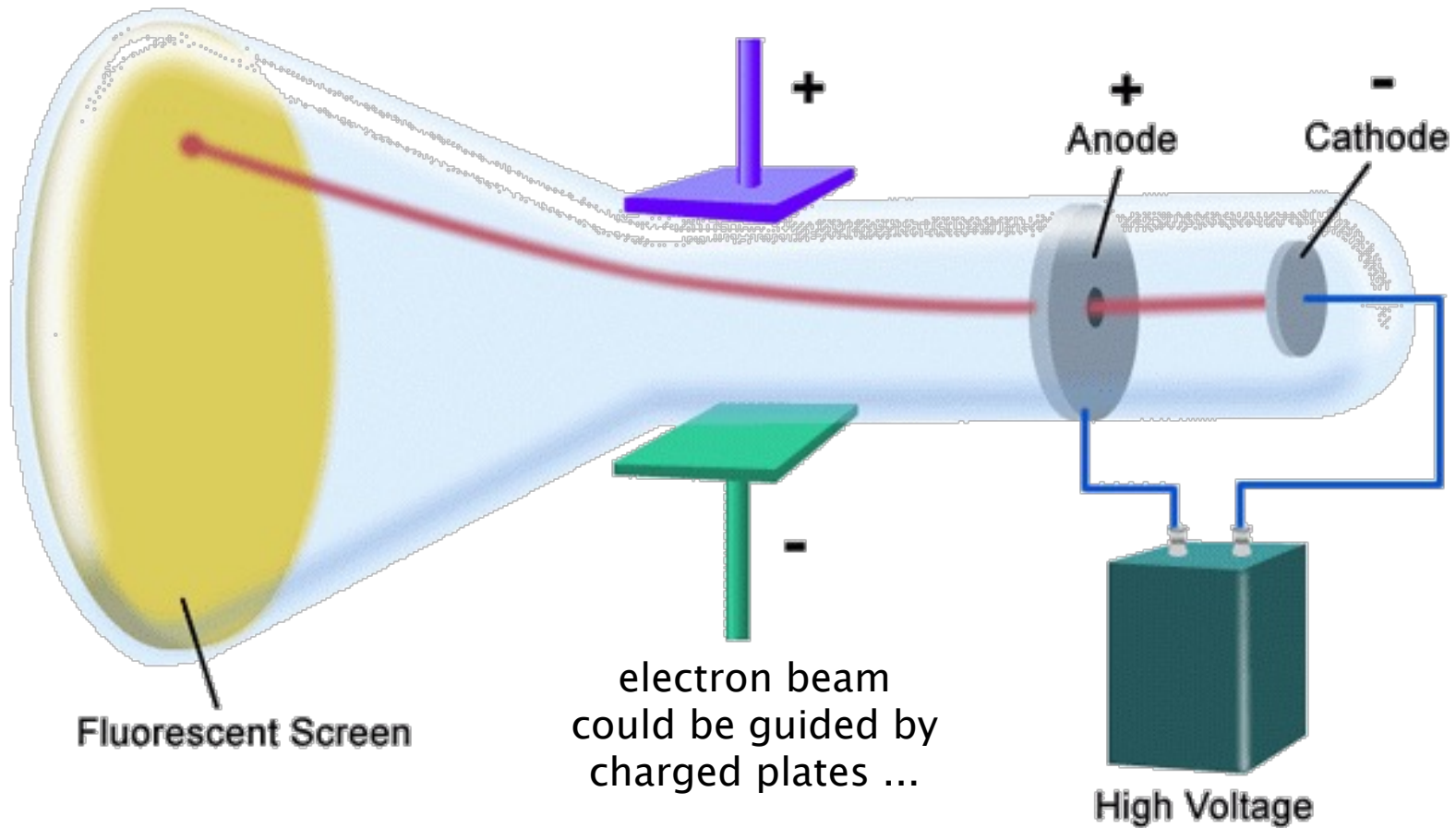


©NCSSM 2002

Tube consists of a valve:

- the cathode emits electrons ("cathode ray")
 - electrons are negative charges
 - attracted to anode
 - shoot through anode
- can be deflected by a magnetic field ...

Cathode Ray Tube – CRT

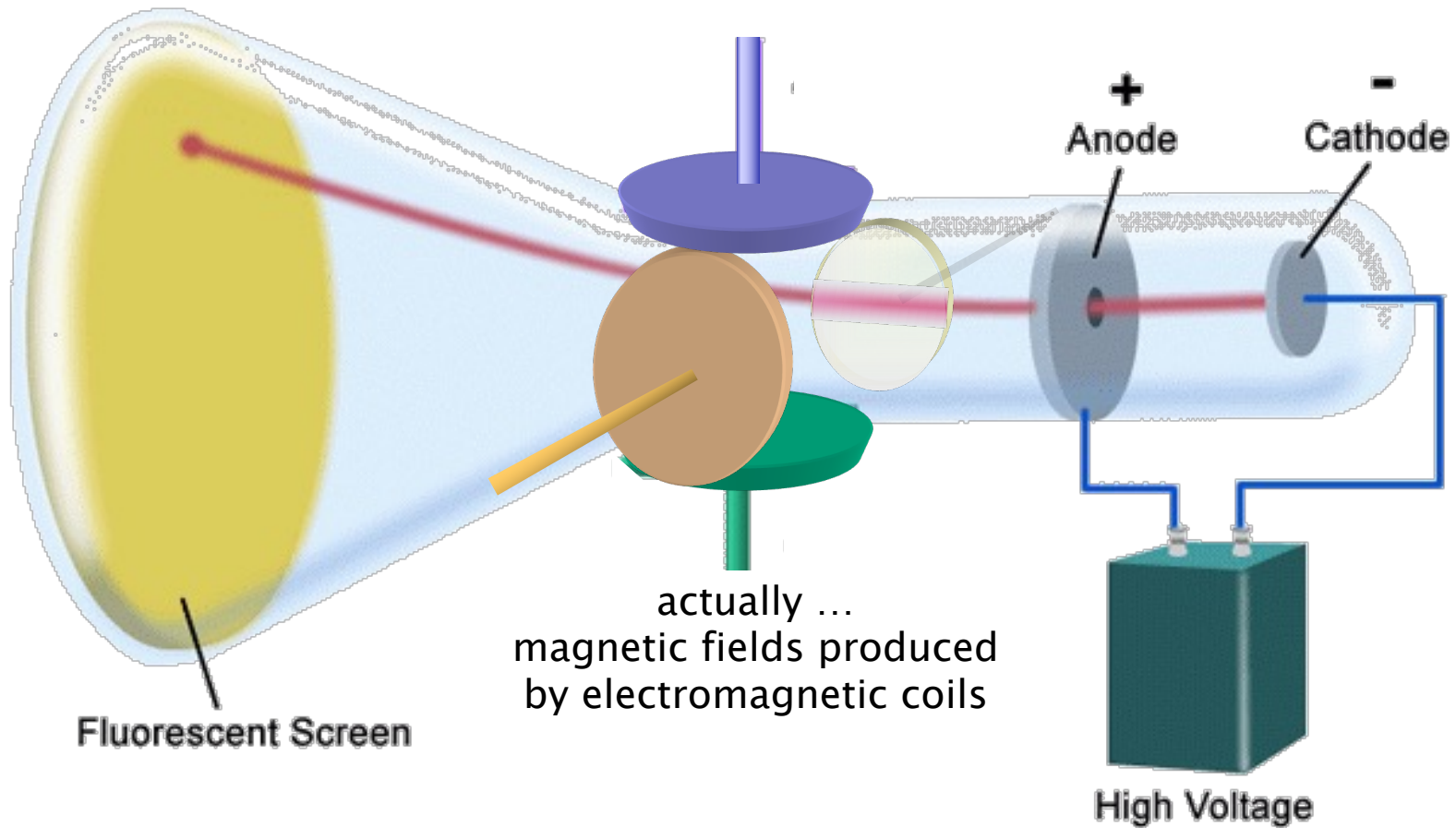


©NCSSM 2002

The fluorescent screen is a coating of phosphor dots which glow when the beam hits them

There are three beams: Red, Blue and Green, to produce RGB pixels on the screen (see later for more details)

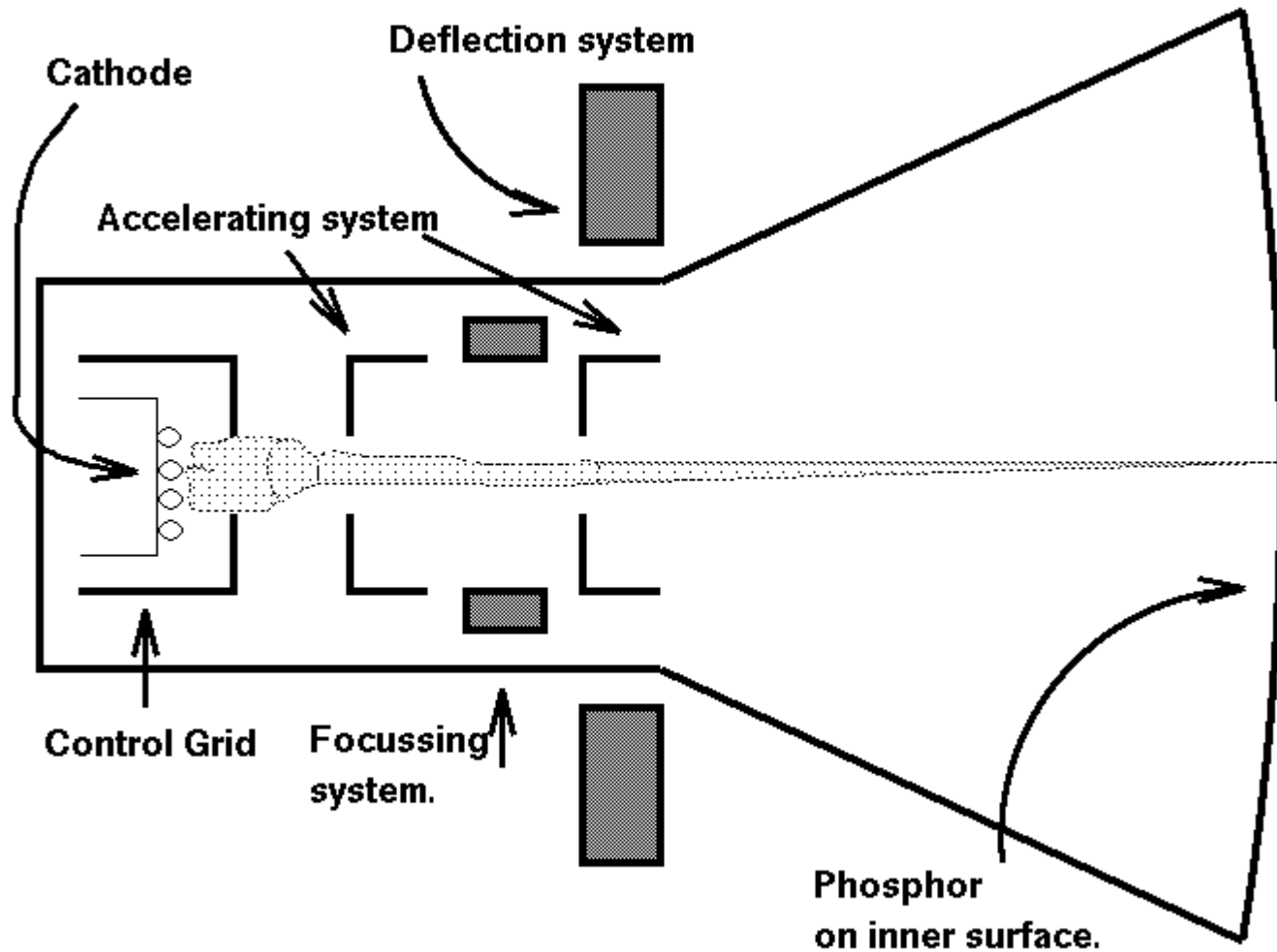
Cathode Ray Tube – CRT



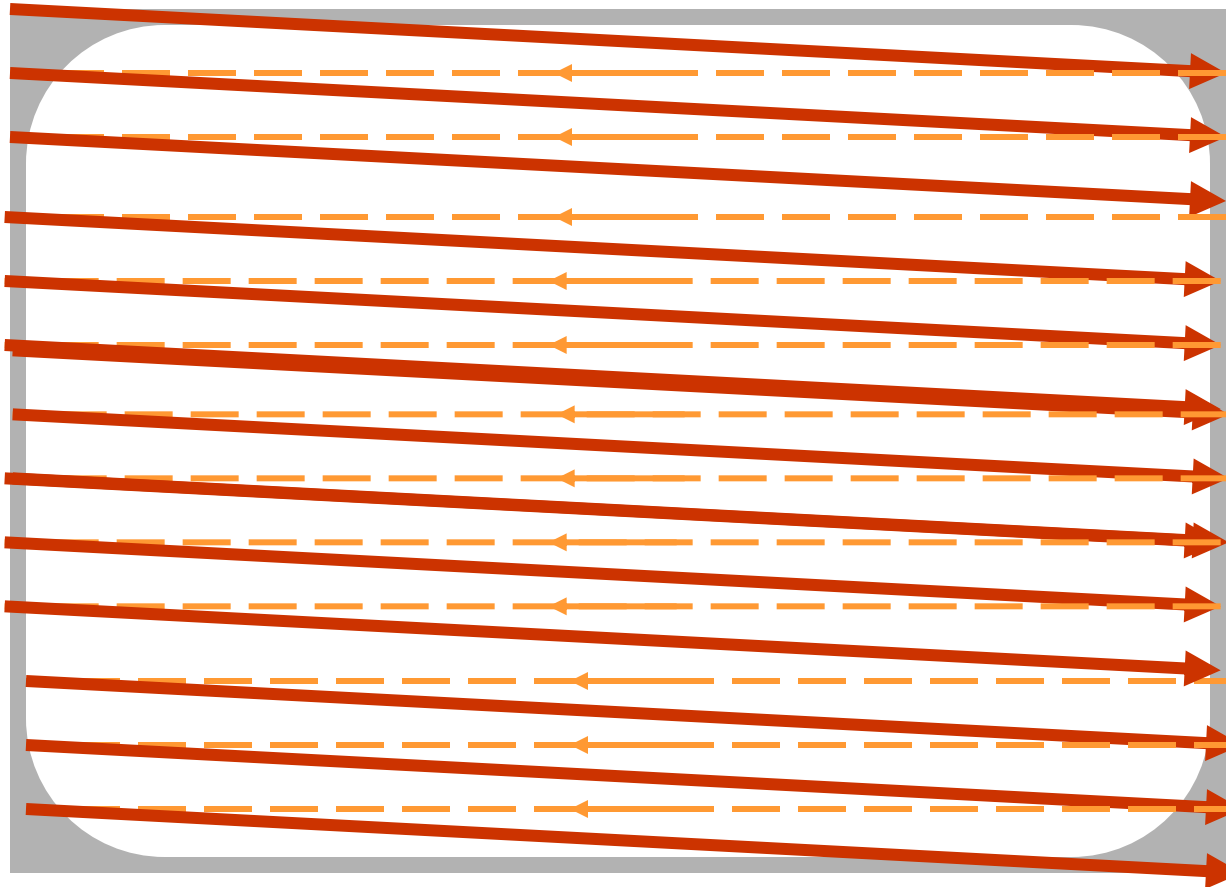
©NCSSM 2002

- Two sets of coils apply appropriate magnetic field alignments to steer the beam vertically and horizontally across the screen
- Voltage to control the coils are derived from the received TV signal

Cathode Ray Tube – CRT

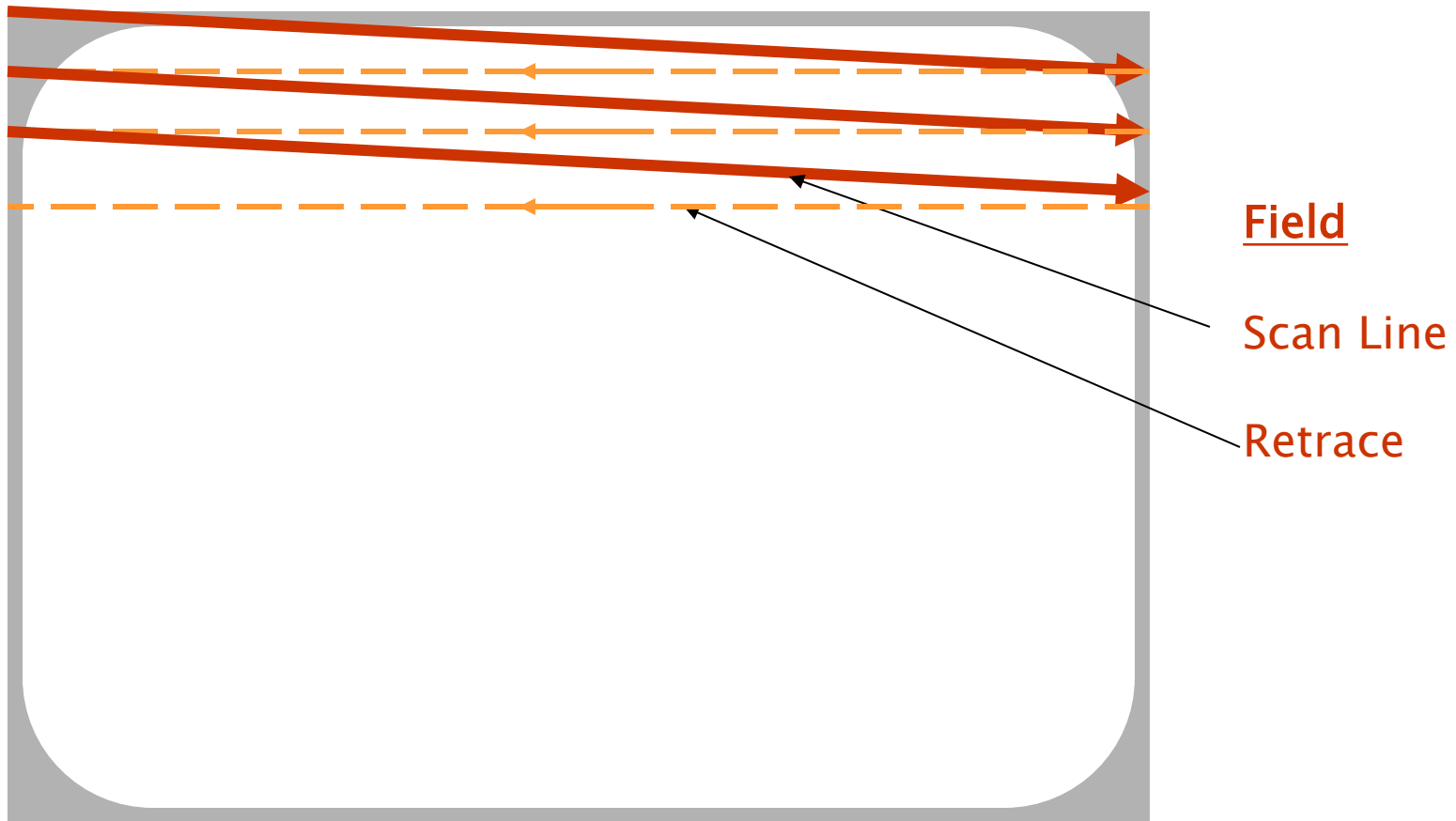


Line Scanning

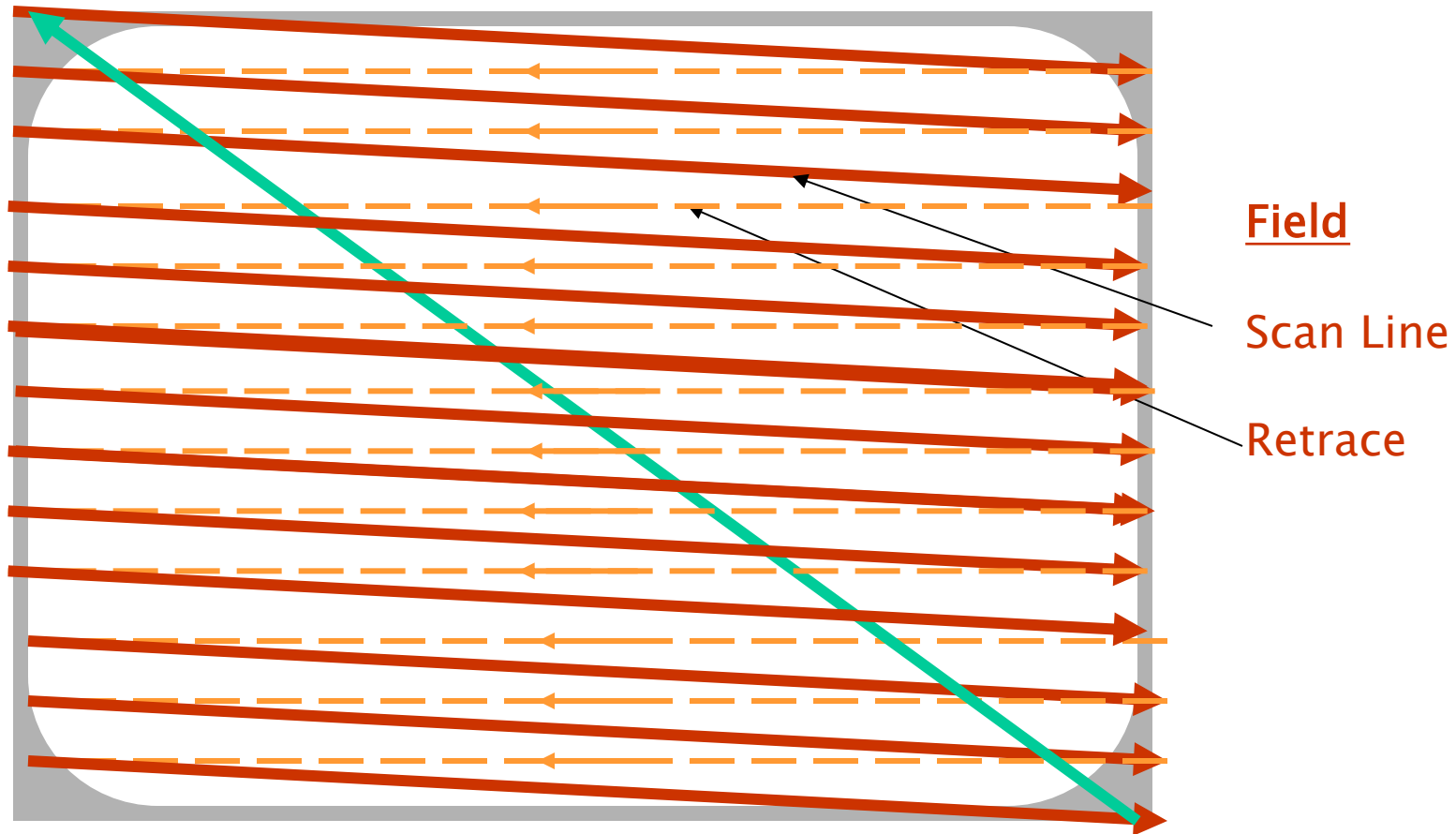


Electron beam is traced across and down the screen, controlled by coils (i.e. electromagnets) fed with a signal: to trace the “across and down” movement of the beam (= 625 lines in PAL)

Progressive Scanning



Progressive Scanning



Progressive Scanning

Movies: 24 full pictorial frames / second

- Flicker
- ... but no motion of the flicker.
- ... eye not so sensitive to flicker in a dark cinema

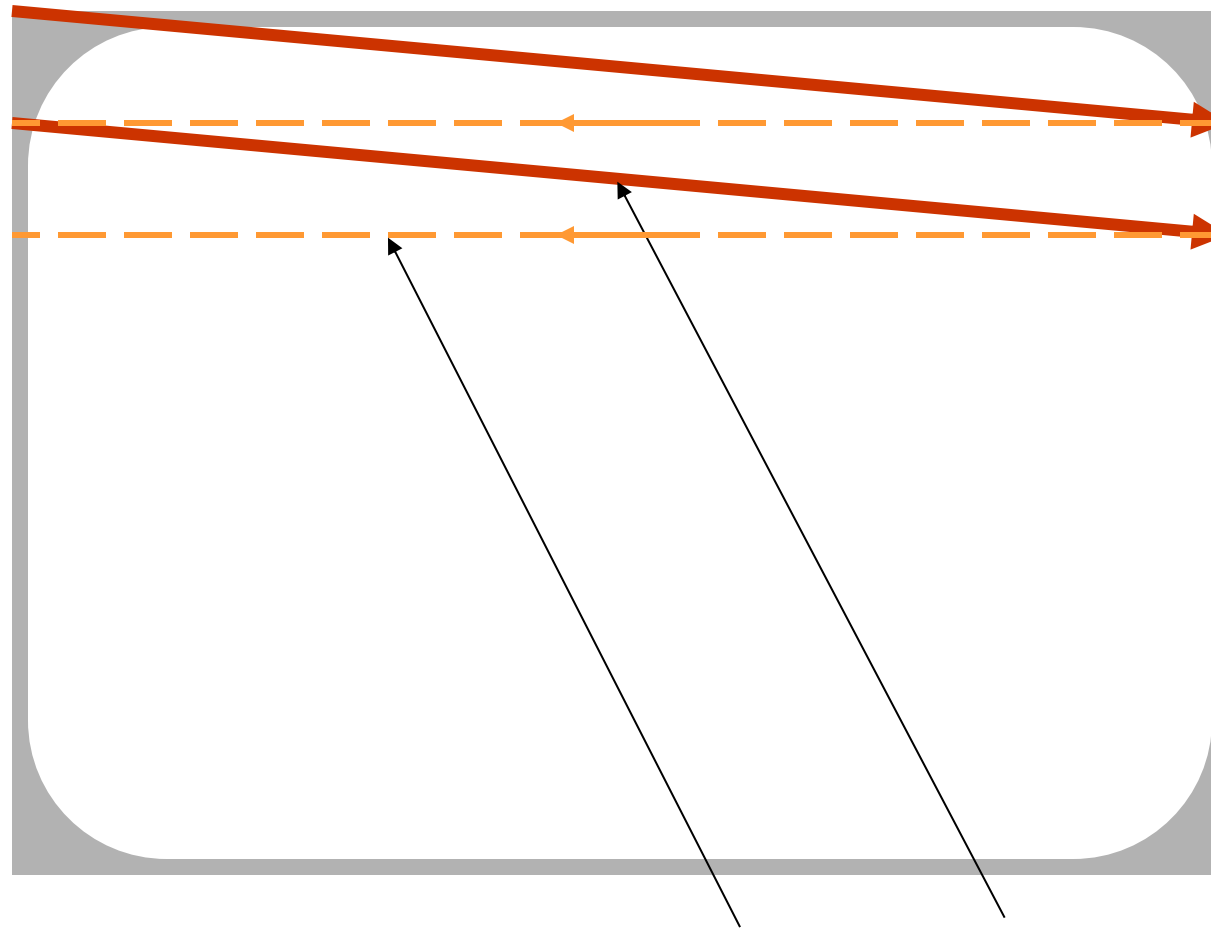
TV

- Scanned – light emission

... Need to reduce flicker without requiring excessive bandwidth

Interlaced Scanning

To give (“apparent”) higher resolution / doubling the frame rate whilst conserving bandwidth



Field 1: Scan Line; Retrace

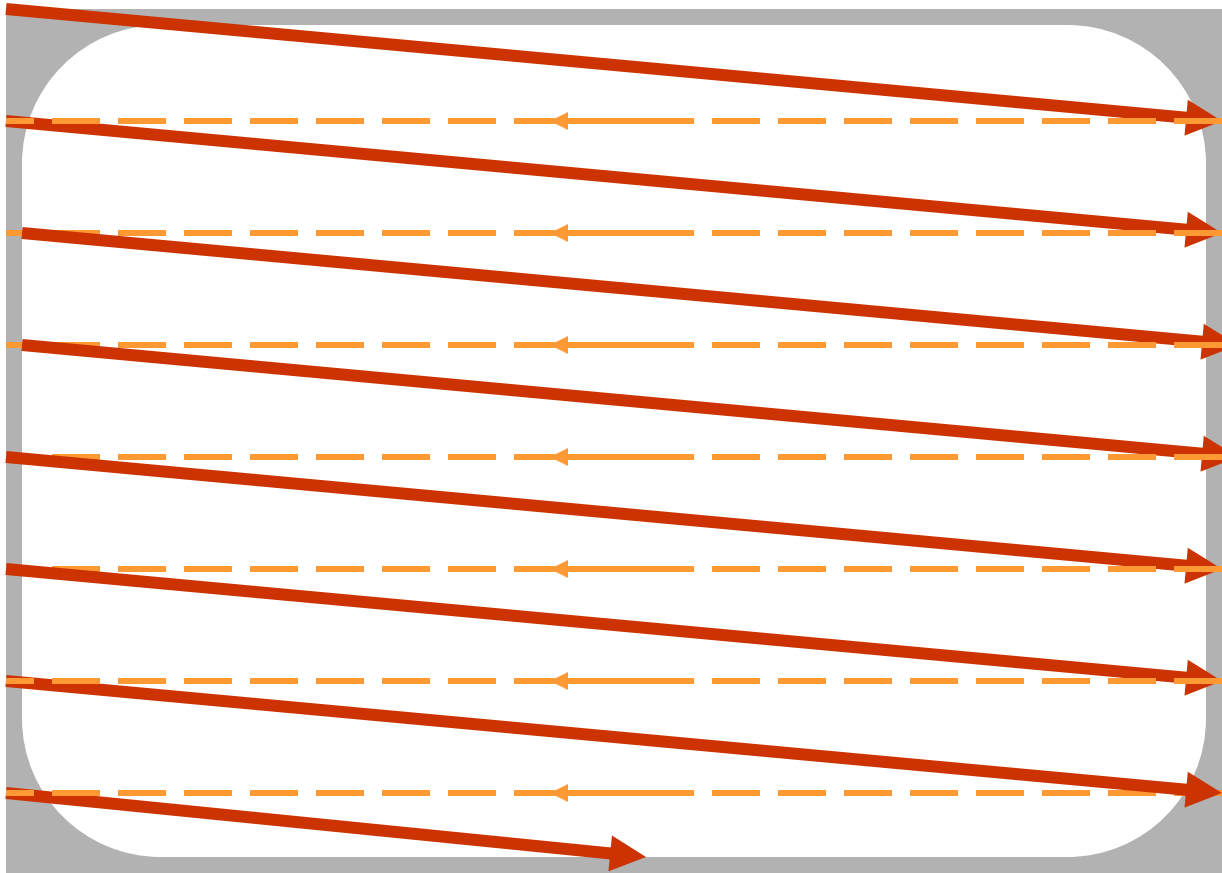
Odd / even lines are scanned alternately –

- Frame consists of two fields, each at 50 Hz (PAL system); ... i.e. 25 full frames / second

- Line scan rate is determined by human eye persistence i.e. at this frequency (and higher) the eye can't detect the difference between scans

Only achievable on CRT, plasma displays

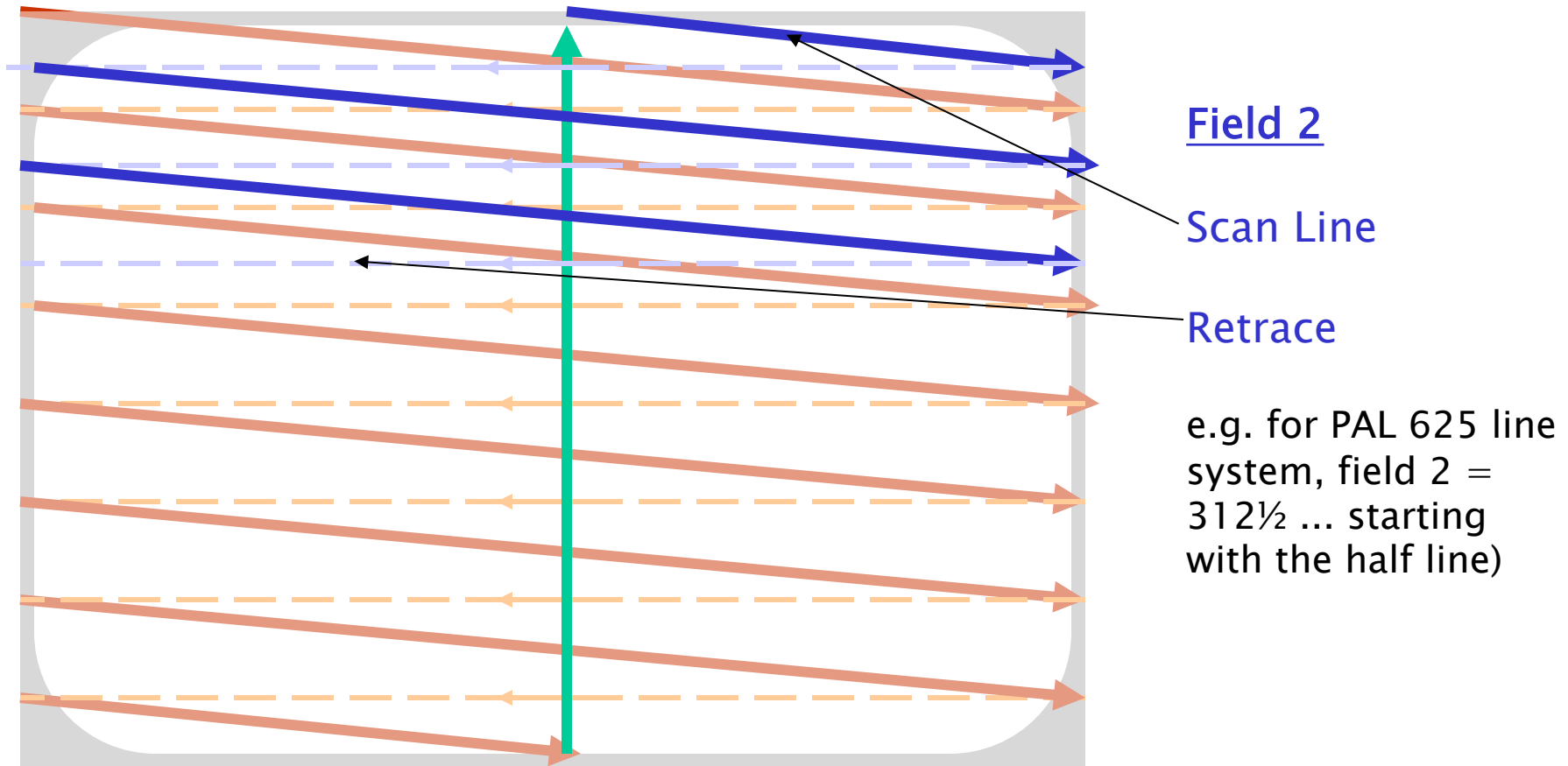
Interlaced Scanning



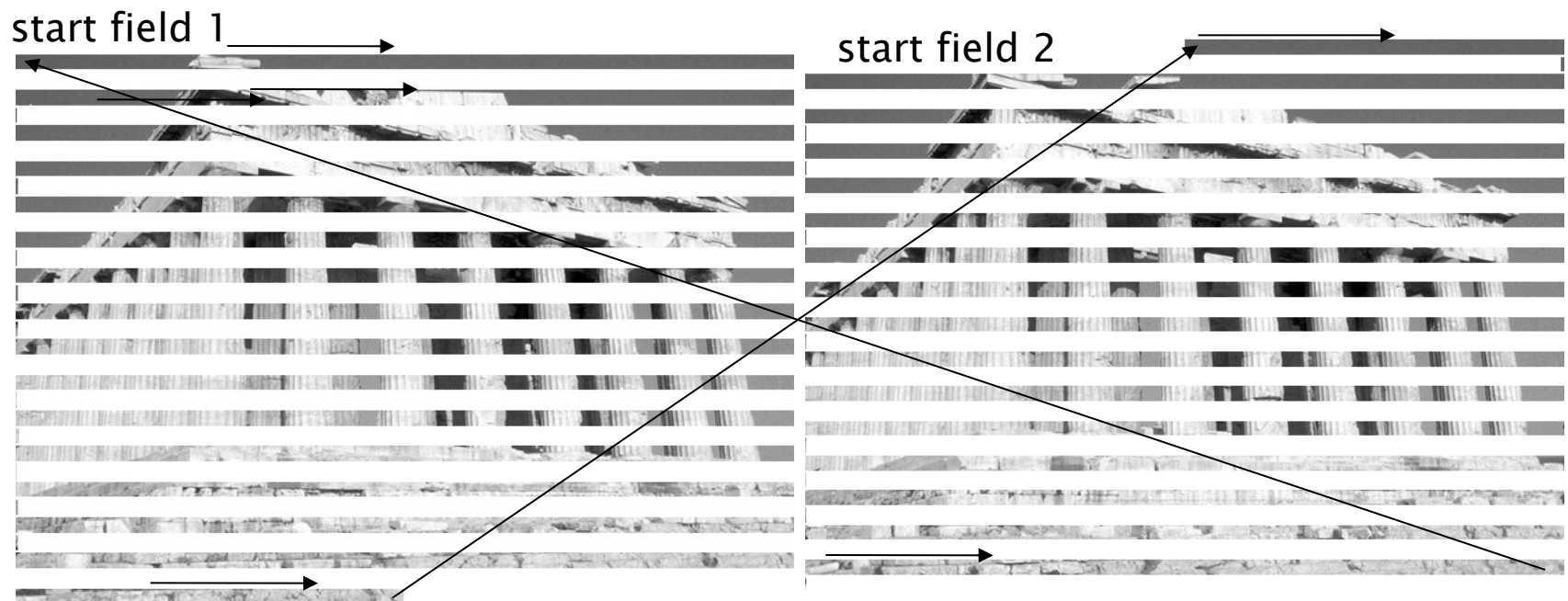
Fully scanned field 1

e.g. for PAL 625 line system, field 1 + 312½ lines

Interlaced Scanning



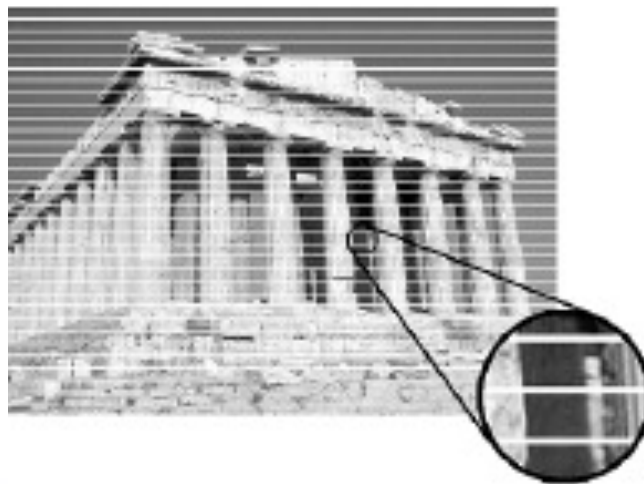
Interlaced Scanning



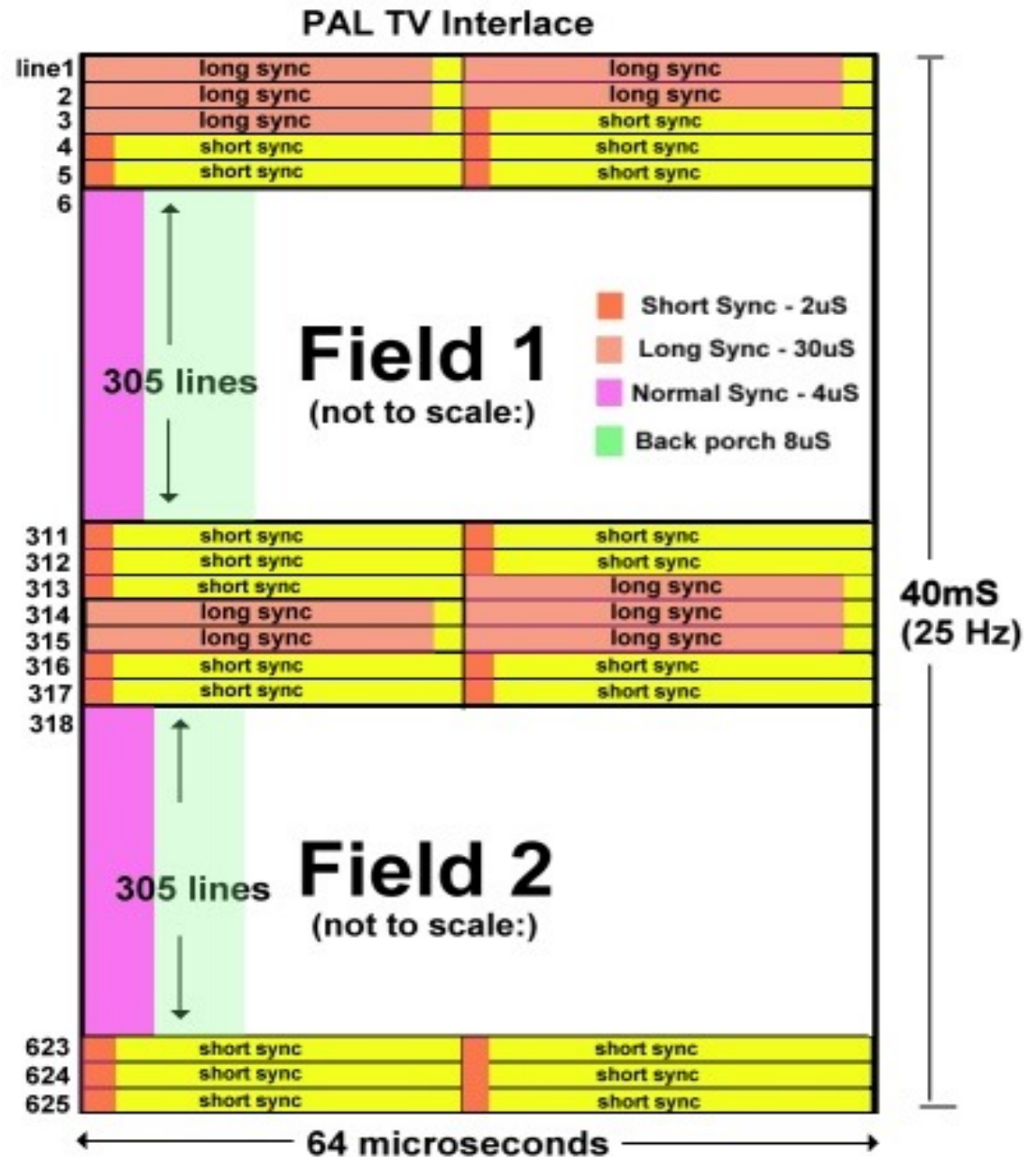
field 1 =
 $1/50^{\text{th}}$ second

field 2 =
 $1/50^{\text{th}}$ second

complete frame
= $1/25^{\text{th}}$ second



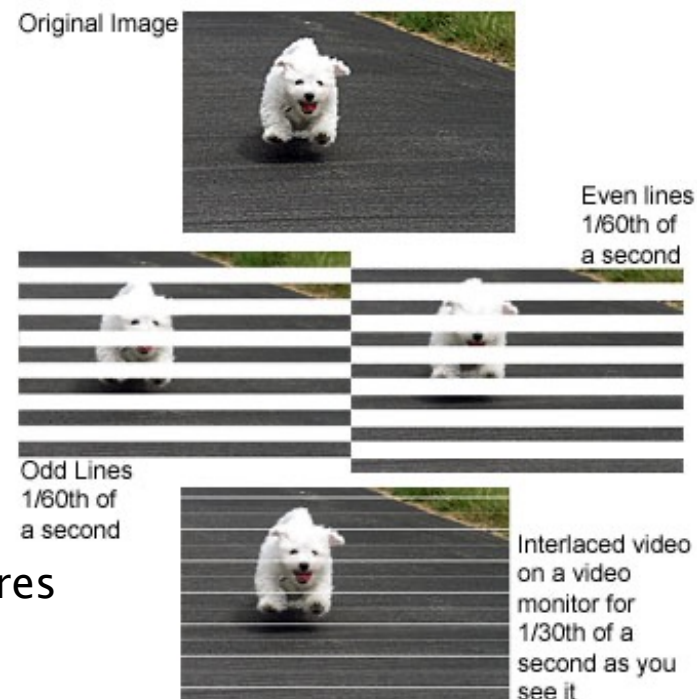
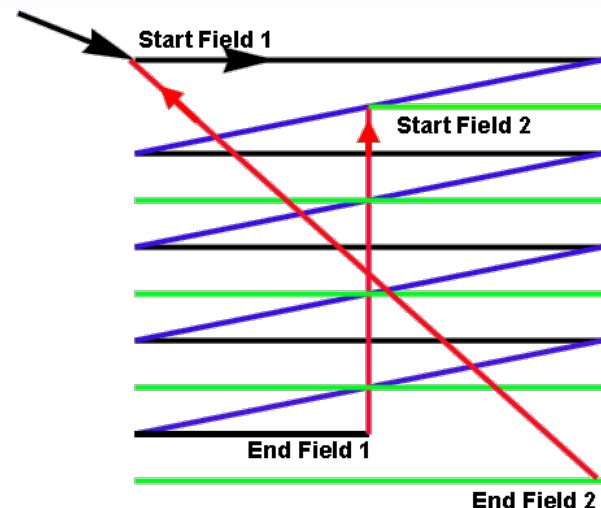
Interlaced Scanning



<http://www.retroleum.co.uk/electronics-articles/pal-tv-timing-and-voltages/>

Frame Rate (In Analogue Days)

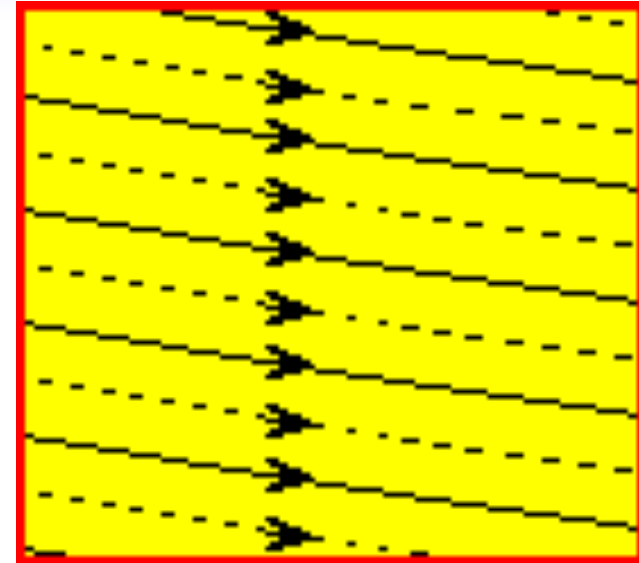
- A Frame represents a complete TV picture
 - analog TV Frame consists of 625 lines.
- Frame = 2 Fields each containing 1/2 the picture information
- PAL system Frame rate = 25 Hz; Field rate is 50 Hz
- NTSC ... 30Hz; Field rate = 60Hz;
- Pictures displayed at frame rate exhibit obvious flicker
- Interleaving the Fields reduces flicker.



NTSC figures

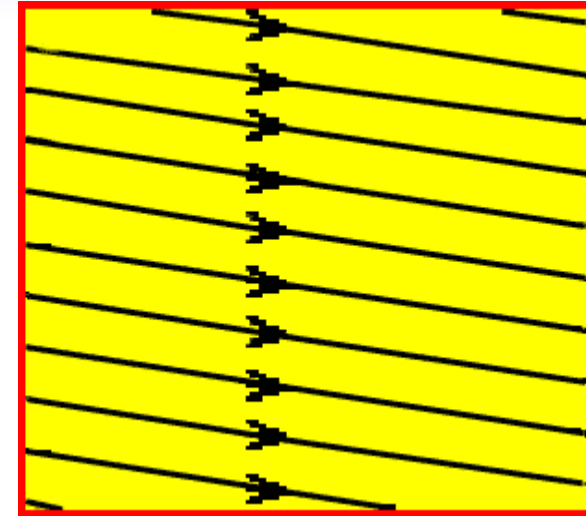
Interlace

- To reduce the perceived screen flicker (25 Hz) on a television, a technique called 'interlacing' is employed.
- Interlacing divides each video frame into two fields; the first field consists of the odd scan lines of the image, and the second field of each frame consists even scan lines.



Progressive Scan

- Simplifies the interpolation and filtering of images
- Allows MPEG-2 compression to work more efficiently by processing complete pictures
- Direct processing of progressively-scanned sources
- 24 frame/second progressive film mode can be provided.
- Assists video conversions with different:
 - numbers of scan lines
 - numbers of samples per line
 - temporal sampling (i.e., picture rate)



(De-) Interlaced Scanning

Problems –

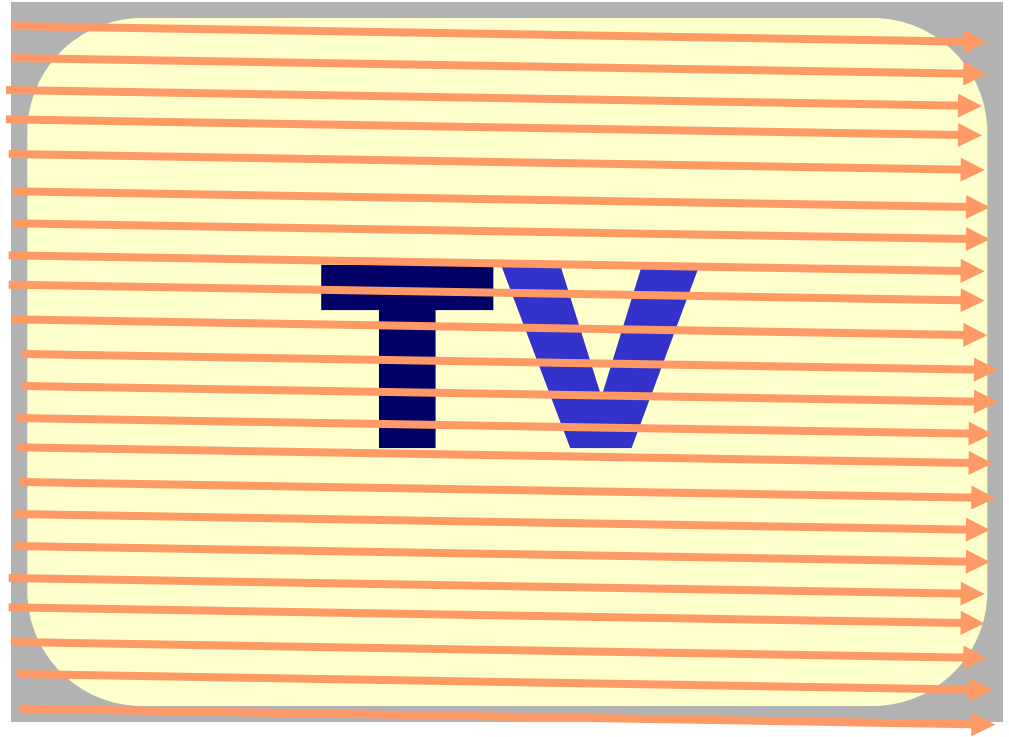
- ... deinterlace required for progressive playouts – LCDs, LEDs, etc.,
- ... introduces lag between frames



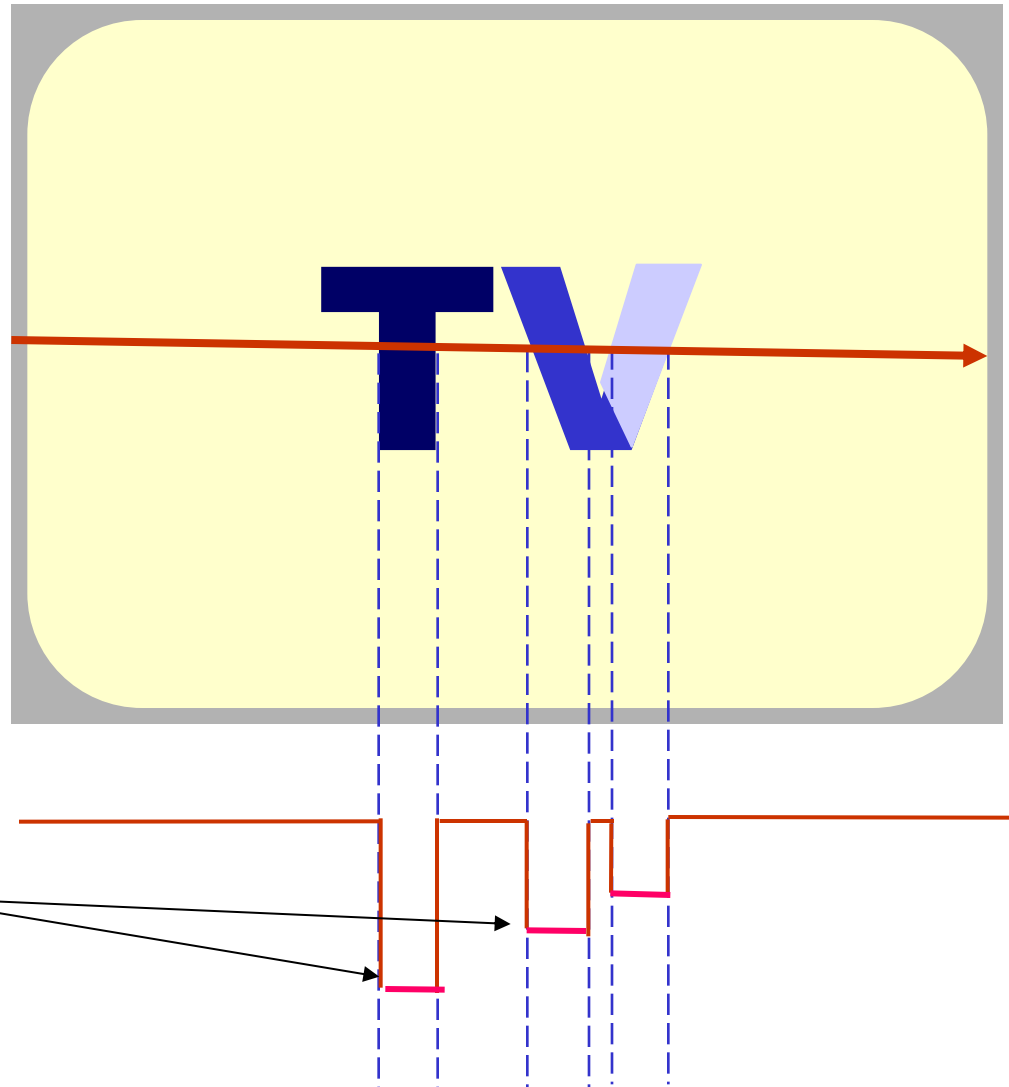
Scan Line

e.g. scan this image:

- 312 ½ lines for the first field of a 625 line scan



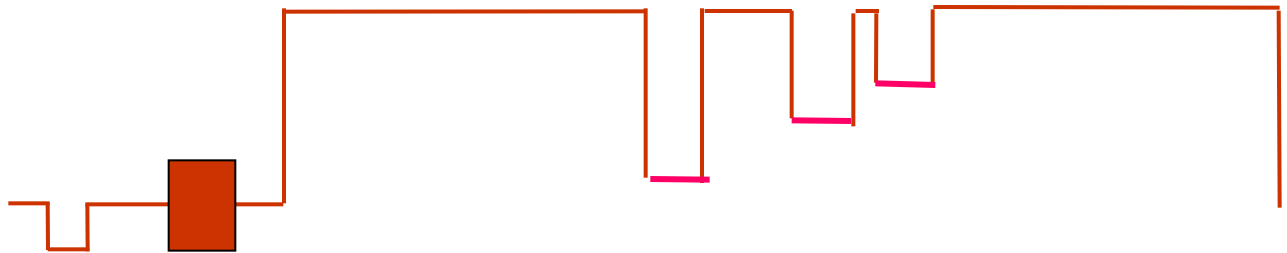
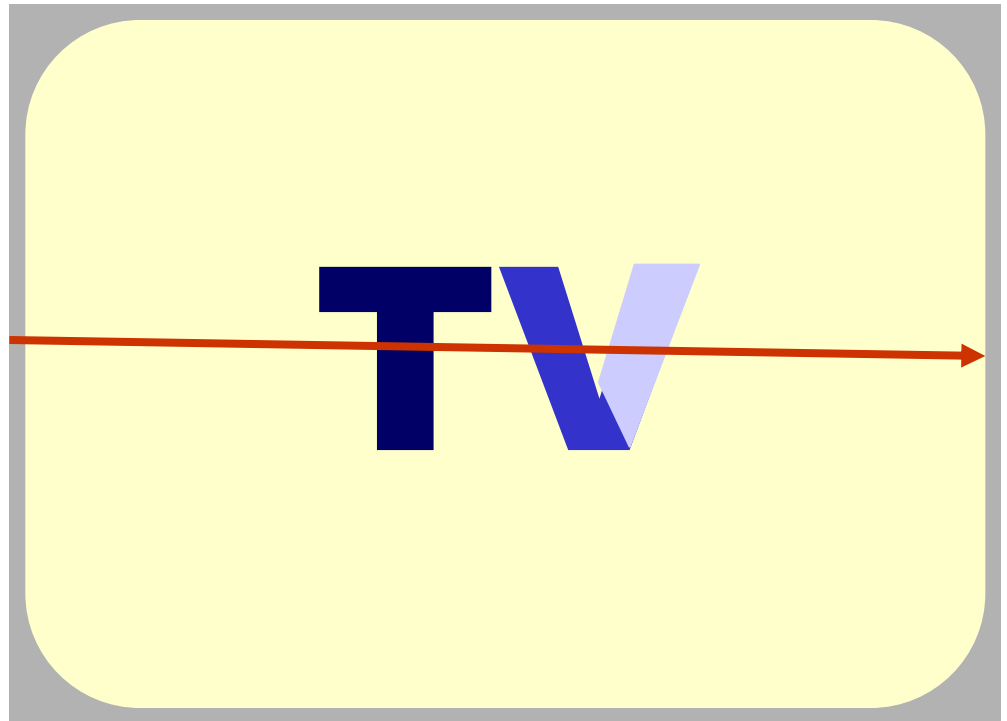
Scan Line – Brightness



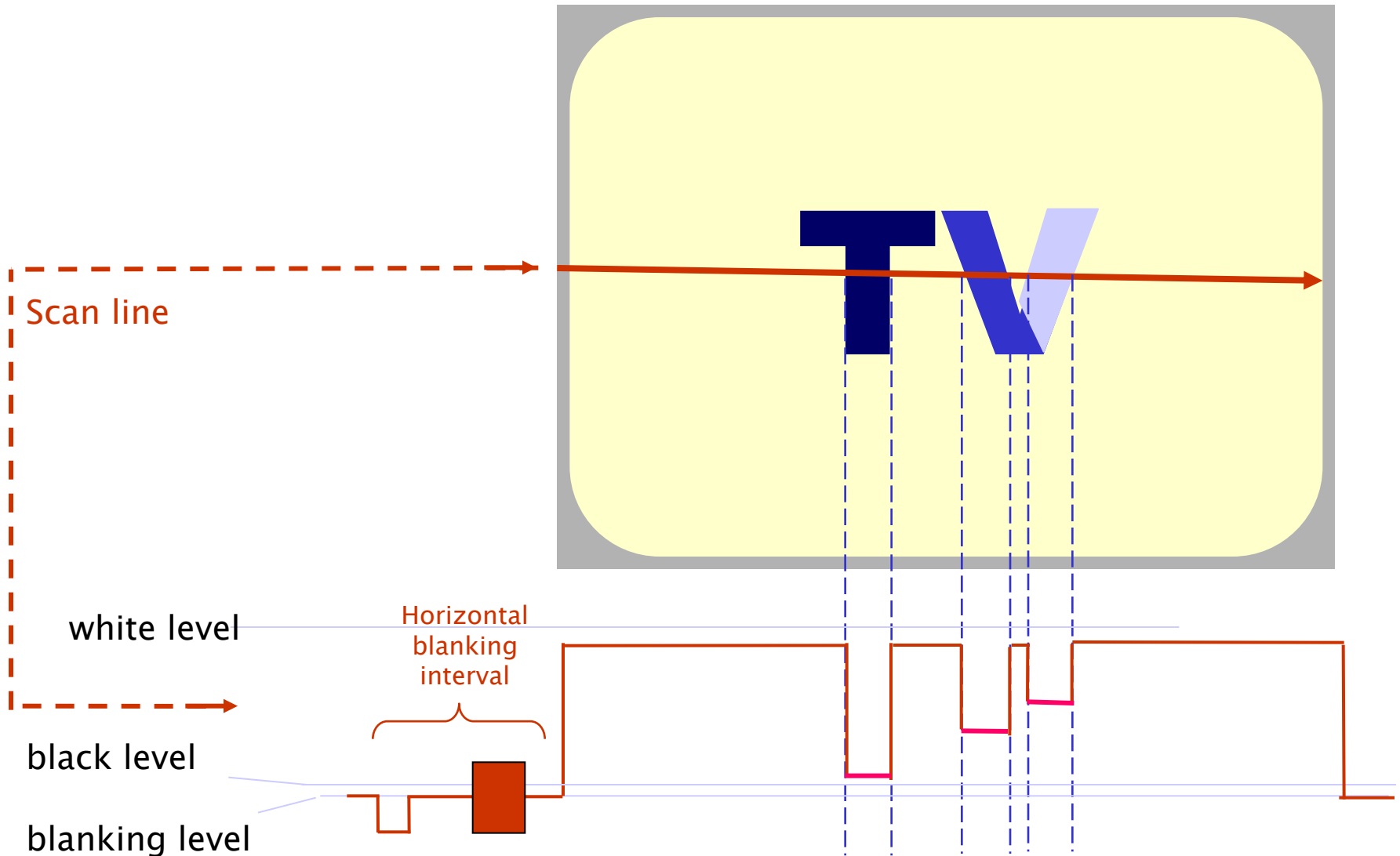
Voltage of signal denotes
the required level of
brightness in the image

Scan Line – Brightness

... now add the line
synchronisation at the
start of the line

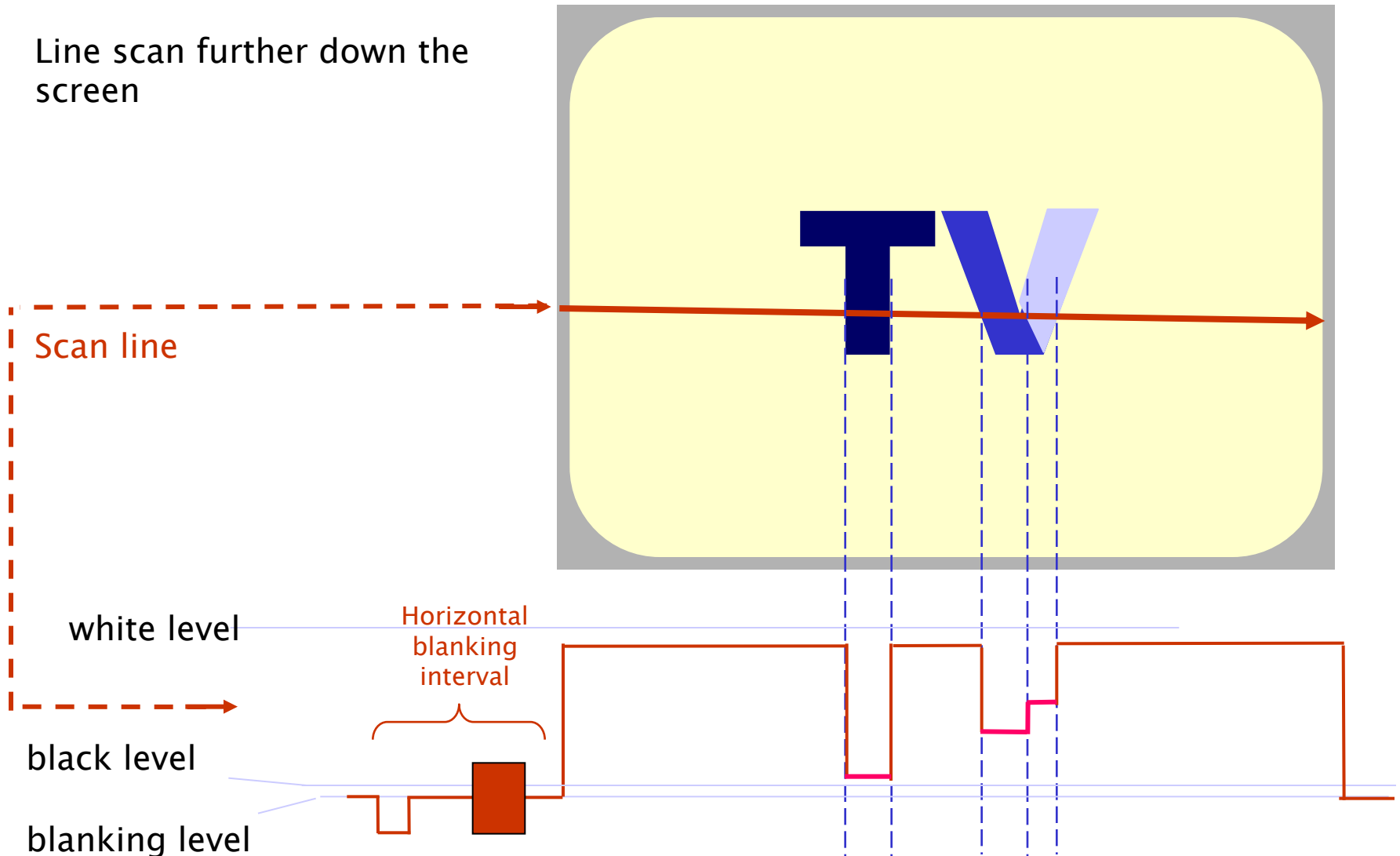


Scan Line – Brightness

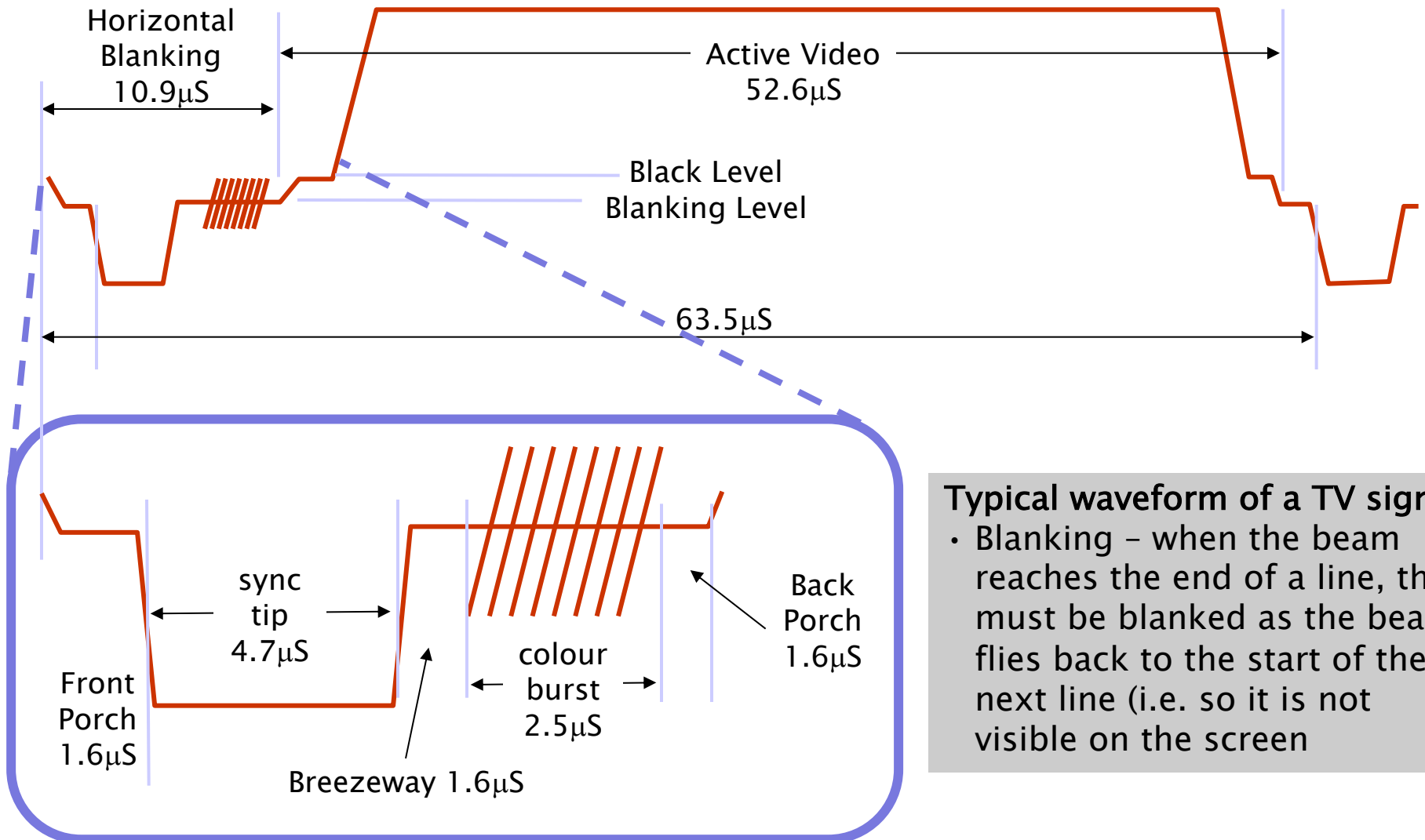


Scan Line – Brightness

Line scan further down the screen



Scan Line Details

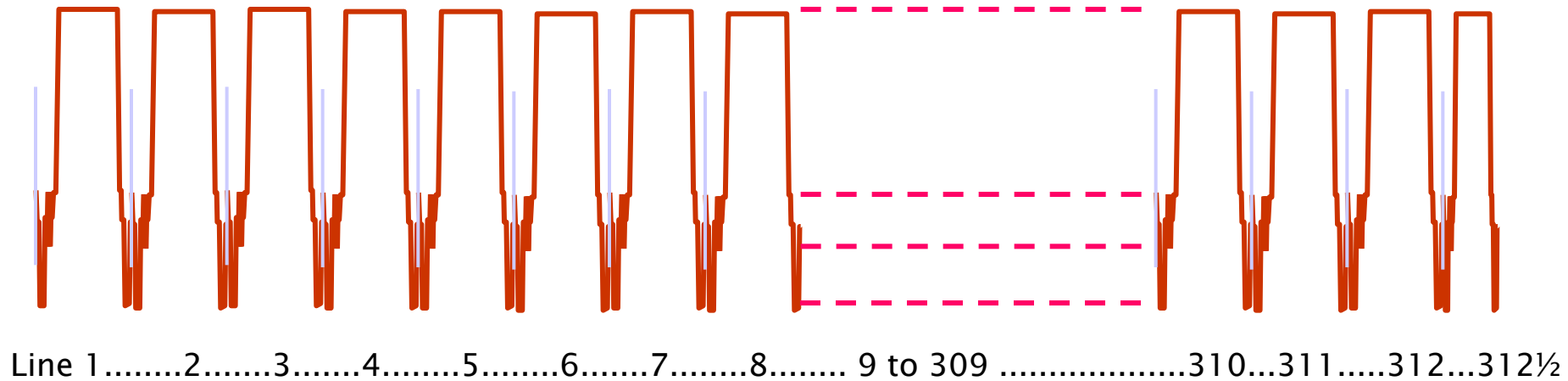


Typical waveform of a TV signal

- Blanking – when the beam reaches the end of a line, the must be blanked as the beam flies back to the start of the next line (i.e. so it is not visible on the screen)

Several Lines Progressing Down The Screen...

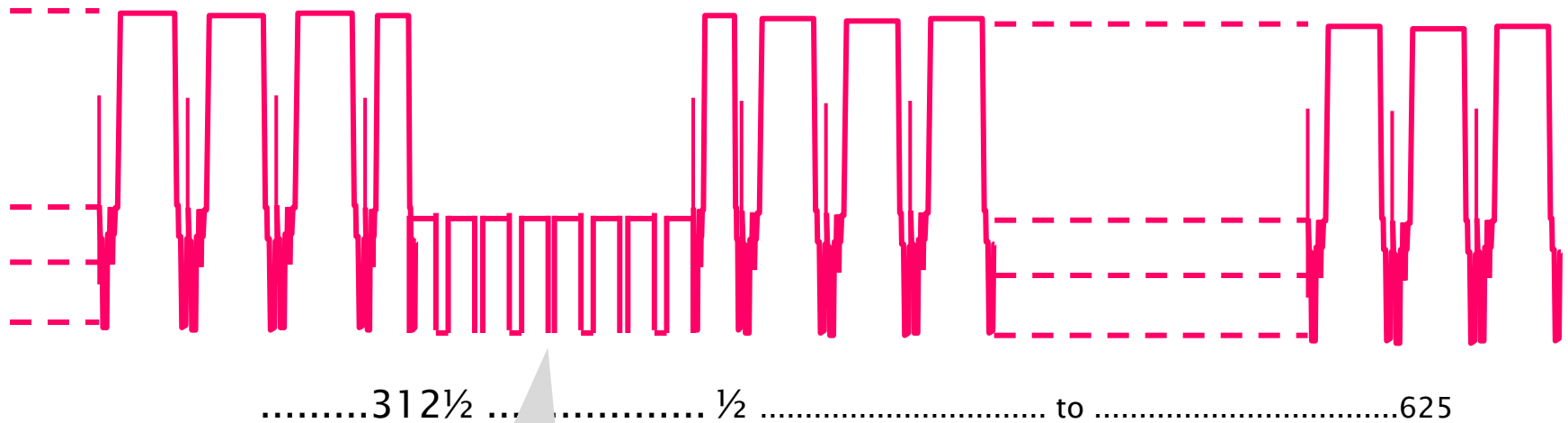
A series of line scans for Field 1:



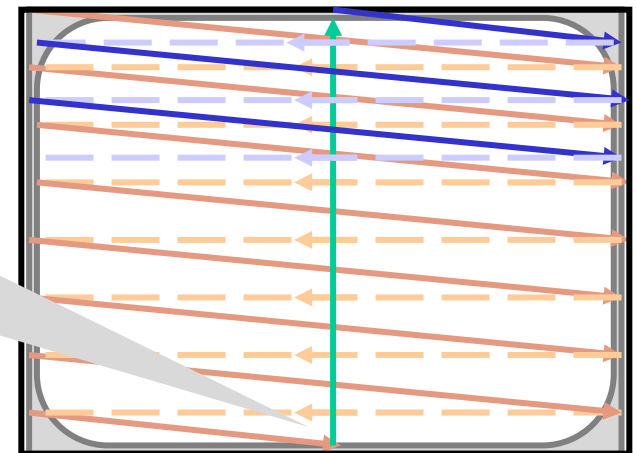
- The scanned lines have now reached the middle of the bottom of the screen
 - hence the half line at the end
- Now they need a sync pulse to get the beam back to the top of the screen for Field 2

Vertical Sync – Return the Beam to the Top of the Screen

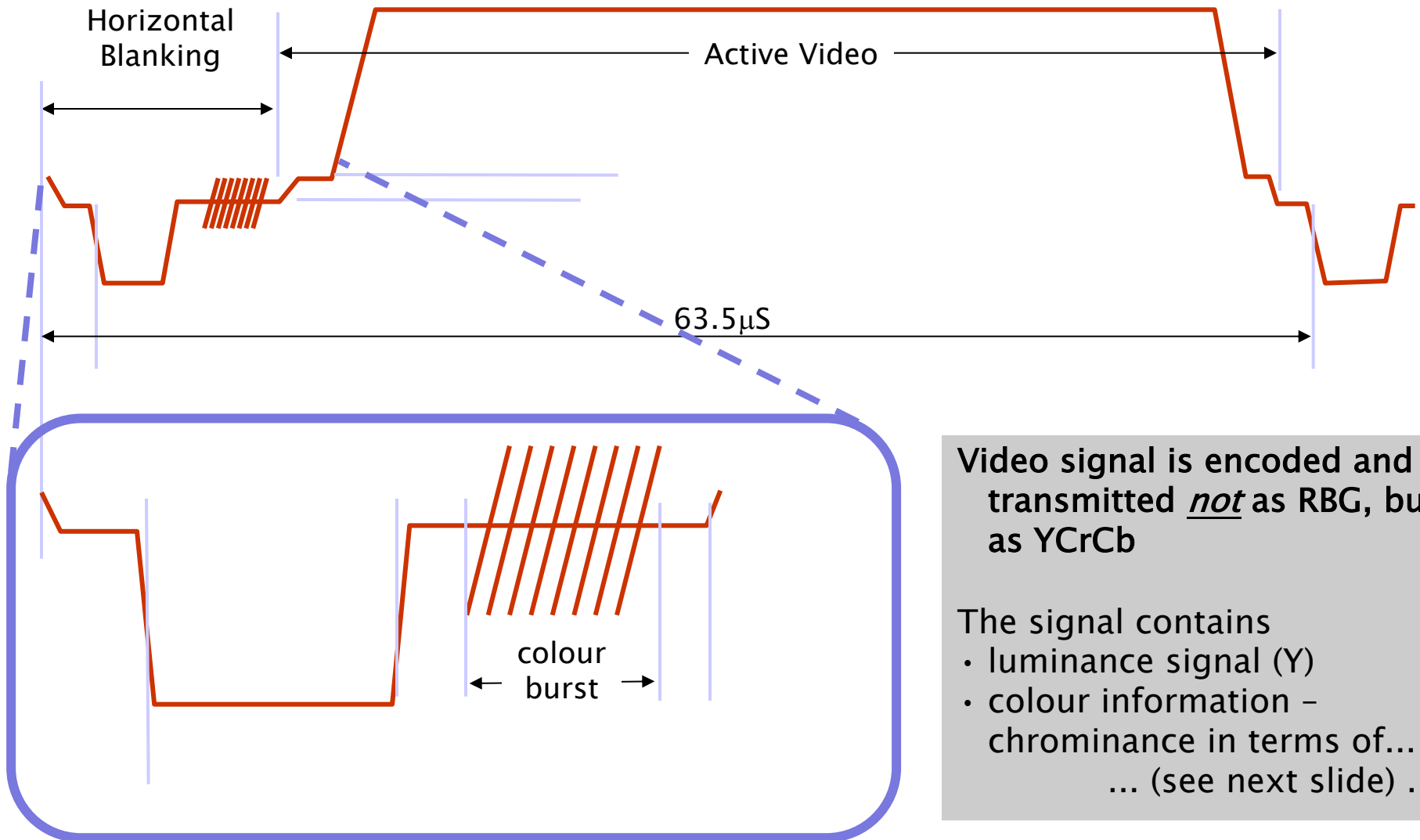
end of Field 1start of Field 2



Vertical sync pulses –
guide beam back to the
middle of the top of the
screen for Field 2



Scan Line Details



Video signal is encoded and transmitted not as RGB, but as YCrCb

The signal contains

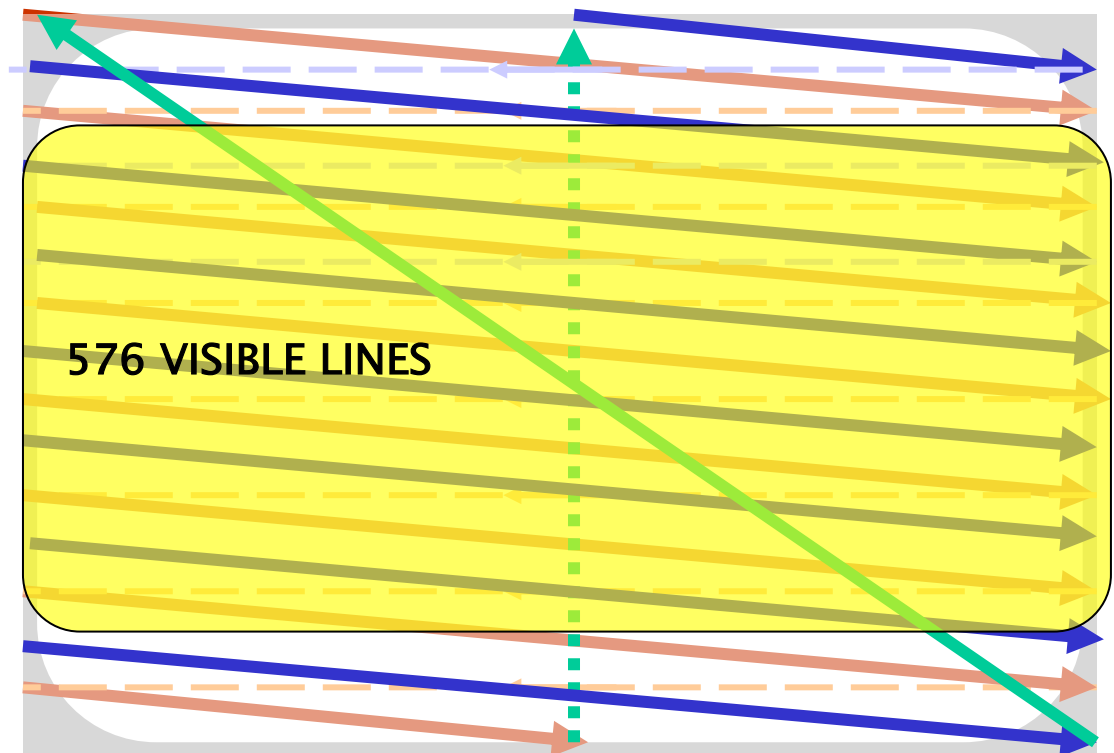
- luminance signal (Y)
- colour information - chrominance in terms of...
... (see next slide) ...

“Visible Lines”

- Vertical retrace is slow compared to horizontal trace (due to magnetic inertia of coils in the receiver)
- The retrace takes the same time as 49 horizontal lines
- History: it is the field governing the coils which control the beam which continues operating whilst the vertical retrace takes place
- This takes the same time as the missing lines

... so only 576 lines carry video (PAL)

- This time is utilised for the horizontal lines to carry additional information ...
 - teletext
 - closed captioning
 - etc



Tri Level Sync – Video Synchronisation, mainly for HD

Syncing devices together

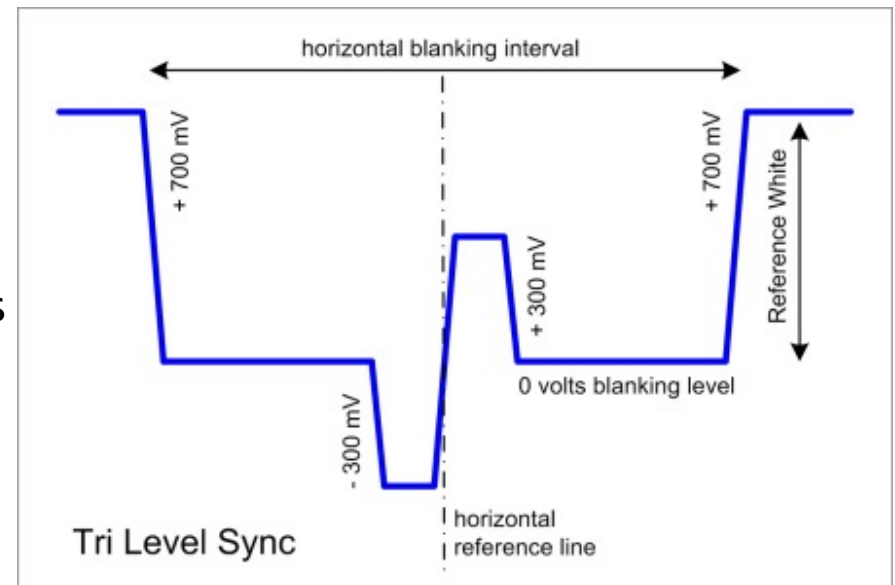
- Genlock (Generator lock)
 - traditionally black-burst (single -ve pulse)
 - vertical, horizontal, frame, and color synchronization.
- HD environments – Tri sync preferred
- timing jitter reduced (higher frequency)
- pulses in both polarities

Used in HD facilities

- equipment output HD-SDI video
- where baseband video needs to be
 - mixed
 - switched
 - luma keyed with any other sources

... need to be synchronous

- first pixel of first line transmitted at the same time
- allows switcher to cut, mix or key sources together
 - ... min delay (~1 HD video line)



<https://www.youtube.com/watch?v=qgmbrt17DiE>

TI:ME:CO;DE

What is Timecode?

Timecode, specifically SMPTE Timecode:

... mechanism for labelling video or film frames
(film refers to timecode as keycode)

Traditionally timecode was recorded by a vtron a dedicated track

- Known as Longitudinal, or linear, timecode

Supported a wide range of shuttle speeds

but not available when the tape wasn't moving

- VITC –Vertical Interval TimeCode

developed to provide timecode in slow speeds and at still frame

Timecode – now – most likely carried as a dedicated metadata item

- often only the start of the media is specifically labelled with timecode

Timecode and Frame Rates

The fundamental timecode “tick” reflects the frame rate of the associated media

Obvious frame rates are:

- 24 fps (Film Based, Progressive)
- 25 fps (Interlaced video, mostly worldwide except USA and Canada)
- 29.97 fps (Interlaced video, mostly USA and Canada)

Other frame rates include

- 23.976 fps
(Film Based, progressive, used when transferring 24 fps film to 29.97 fps video)
- 50 fps
(Progressive video, mostly worldwide except USA and Canada)
- 59.94 fps
(Progressive video, mostly USA and Canada)

25 and 50 fps timecodes maintain a direct relationship between frame count and the timecode is displayed as HH:MM:SS:FF

At 25 fps there are 2160000 frames in a 24 hour period, time code
00:00:00:00 to 23:59:59:24

The Dreaded Drop Frame

Originally TV in the USA and Canada operated at 30 fps

It was convenient, indeed essential, to maintain a direct relationship between the video scan rate and mains frequency

When colour was introduced in the USA (NTSC) adjacent TV channels resulted in an unacceptable level of interference patterning, particularly on legacy monochrome receivers

The frame rate was offset to 29.97 fps to alleviate the interference

The Dreaded Drop Frame

30 fps has 2592000 frames in 24 hours

29.97 fps has only 2589408 frames in 24 hours

This means that a direct conversion of “frames to timecode” would result in the displayed timecode being too low by some 86 seconds

To compensate for this “Drop Frame” timecode was introduced

- However no *frames* are dropped, only the *frame labels*

Drop frame timecode omits certain timecodes on a predictable basis

- First two frames XX:XX:XX:00 and XX:XX:XX:01 of each minute

except

when the minute is exactly divisible by 10

(minutes MOD 10 = 0)

Drop Frame Timecode Example

01:08:59;28

01:08:59;29

01:09:00;02

01:09:00;03

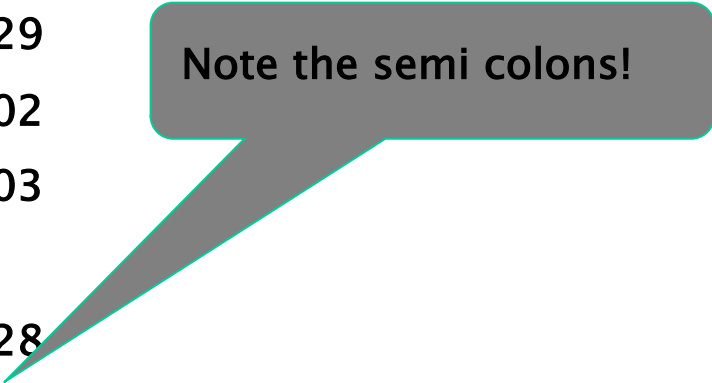
...

01:09:59;28

01:09:59;29

01:10:00;00

01:10:00;01



Note the semi colons!

Progressive Timecode

Because the original timecode format never envisaged 50 fps and 59.94 fps some systems are unable to display progressive timecodes correctly

These systems typically share the same timecode across two frames

The “frames colon” may be used to indicate which frame of the pair is displayed

50 fps display	25 fps display
00:10:05:00	00:10:05.00
00:10:05:01	00:10:05:00
00:10:05:02	00:10:05.01



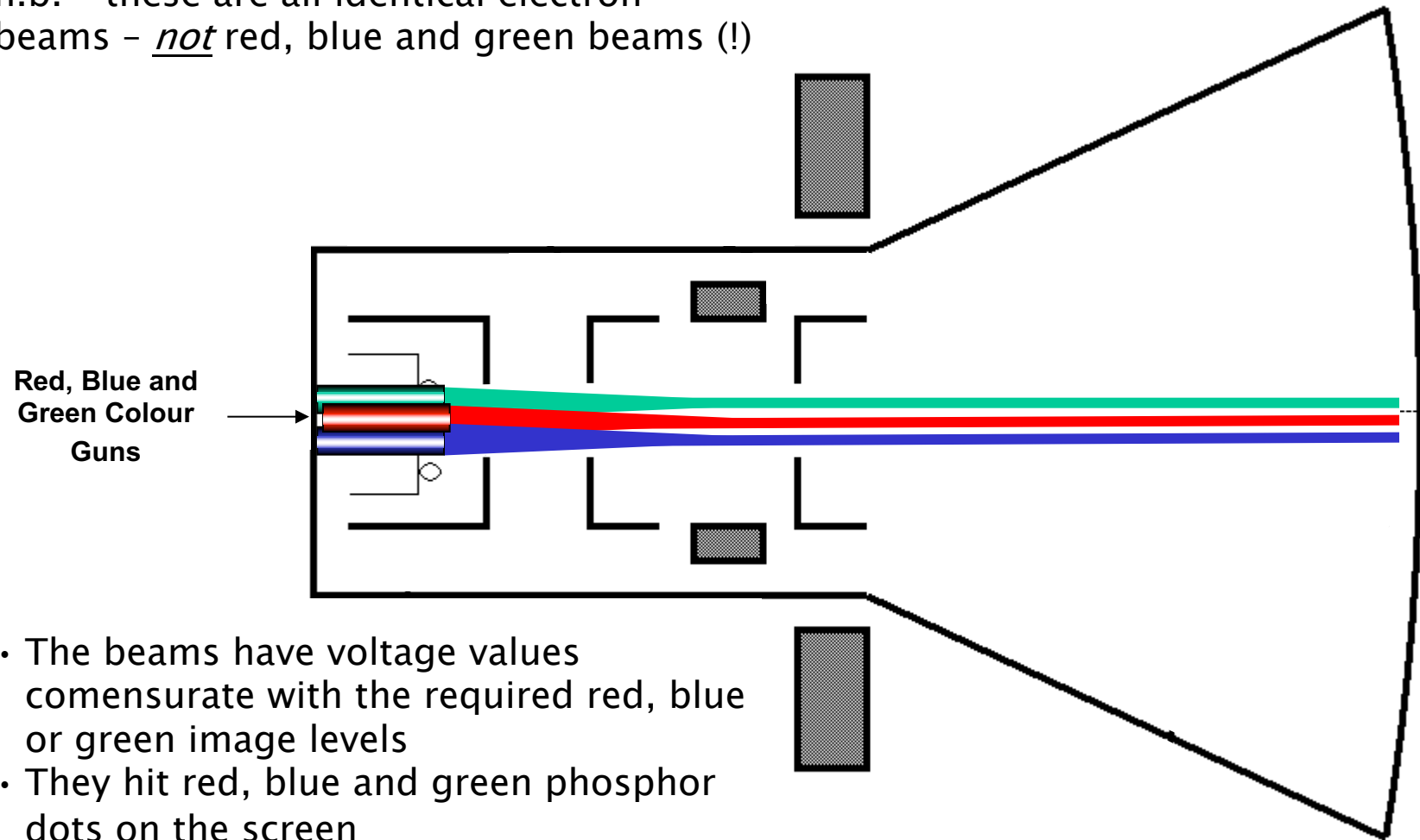
B B C HD

1080 lines

Colour

Colour – Requires 3 Beams – R, G, B

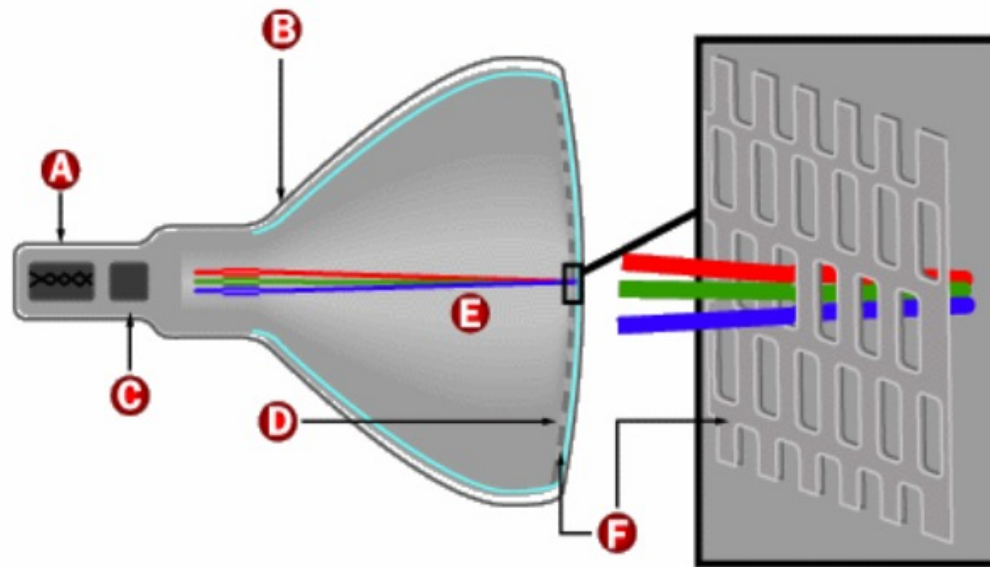
n.b. – these are all identical electron beams – not red, blue and green beams (!)



- The beams have voltage values comensurate with the required red, blue or green image levels
- They hit red, blue and green phosphor dots on the screen

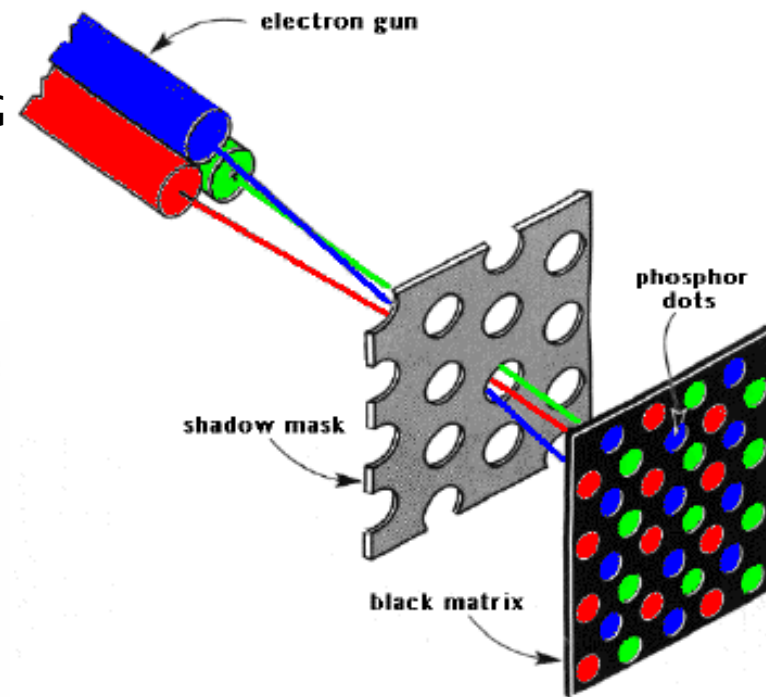
Colour Scanning via Shadowmask

- Red Blue and Green colour guns scan image on to screen
... but need to be directed towards specific R, G or B phosphor dots on the screen
- A shadowmask masks off the areas each beam should not cover



A Cathode
B Conductive coating
C Anode

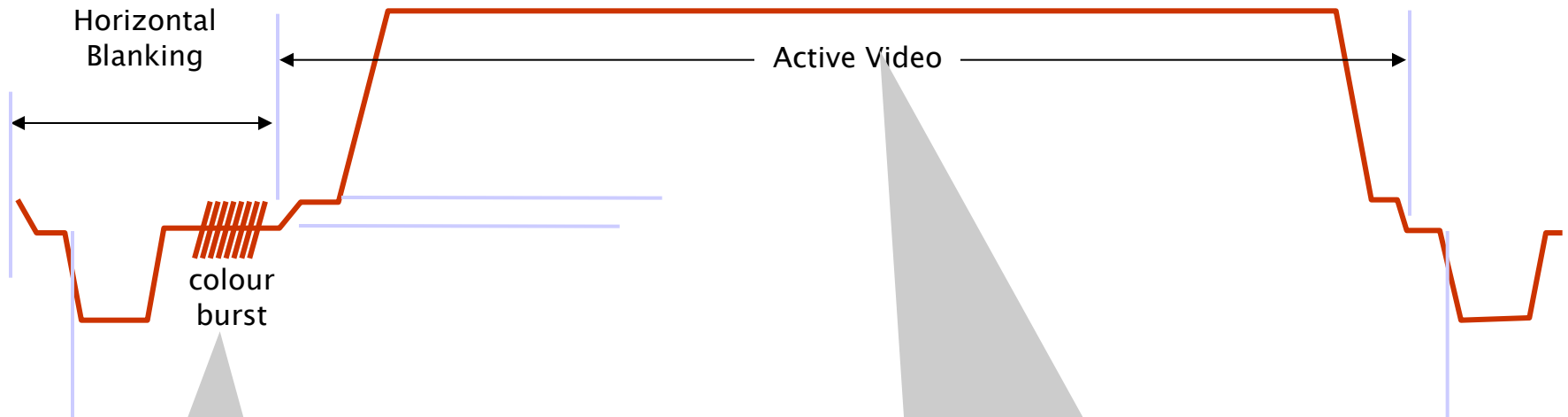
D Phosphor-coated screen
E Electron beams
F Shadow mask



An alternative to the shadowmask is an aperture grille

- similar function
- allows flatter displays

Decoding Colour Information (analogue scans)



Colour burst
gives information on
synchronising the
demodulation of the colour
information from this part of
the signal (on a per line basis)

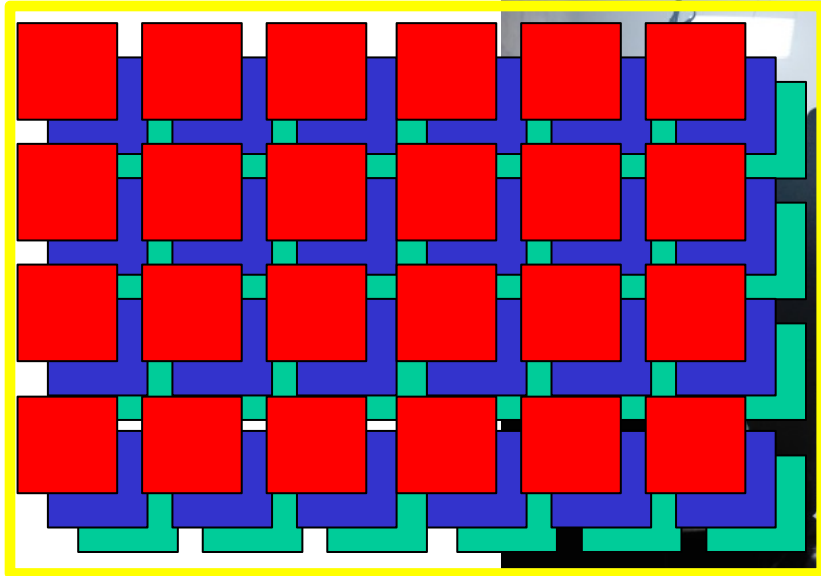
**Active Video section of the signal
(above)**

Y, Cr and Cb (analogue: Y, Pb,Pr)
signals are modulated on to the
signal carriers:

- FM carriers + phase modulated
carriers for colour
(chrominance) information

Image Comprises RGB

Original captured
image at camera
sensor:



Each pixel is represented by R, B, G
(each pixel has its own unique value)

Psychovisual – how we perceive images

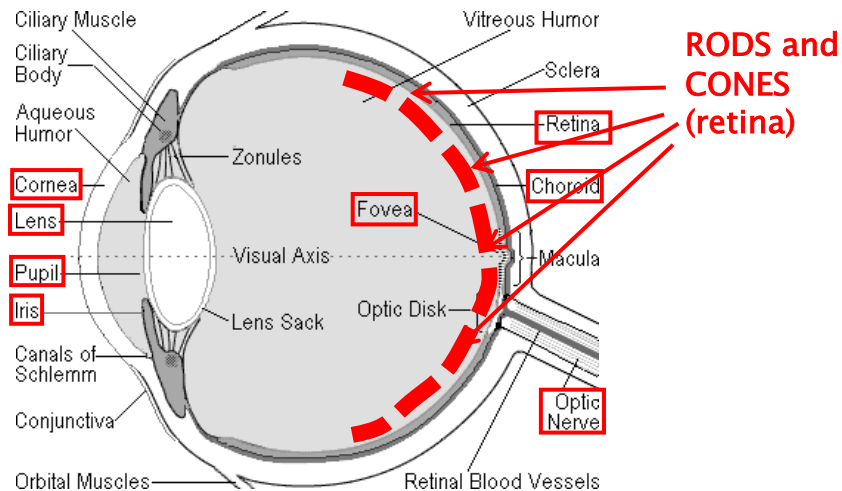
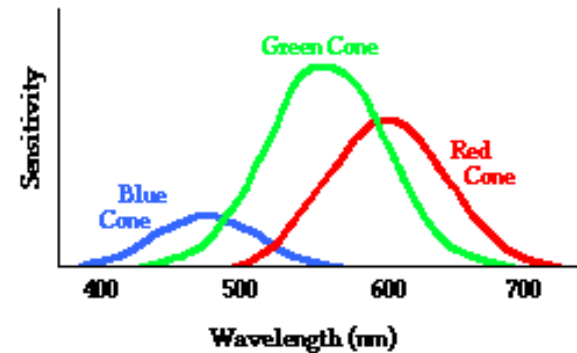
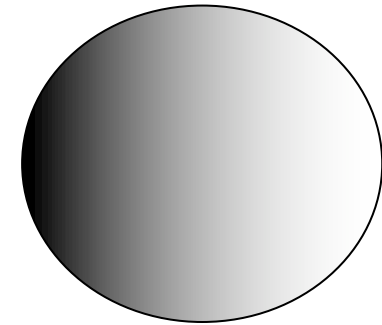
Light sensitivity in the eye

Rods – light intensity – not colour

- ~125 million across the retina
- characteristics of high speed black & white film

Cones – colour – responsive in three wavebands

- 6 or 7 million cones
- characteristics of low-speed colour film



... Chroma Sub-sampling

Colour Image using YCrCb

YCbCr elements of a colour image (digital codings)



Y



R



R

-



Y

=



Cr



G



B

-



Y

=



Cb



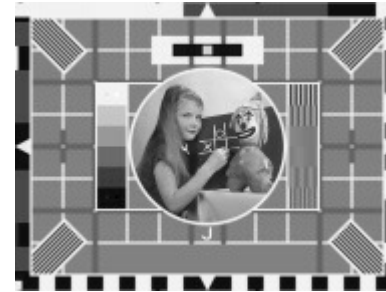
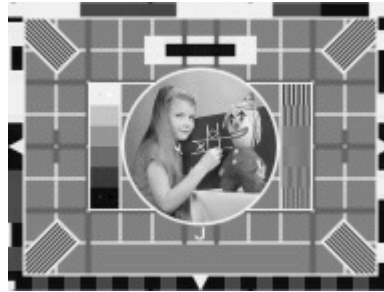
B

i.e. each pixel (RGB) $\rightarrow$ Y, Cb, Cr arithmetic

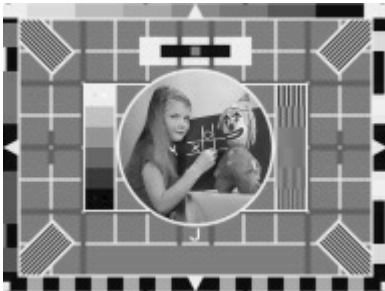
R, G, B and Colour Difference



Original colour



Colour Separation signals R, G, B

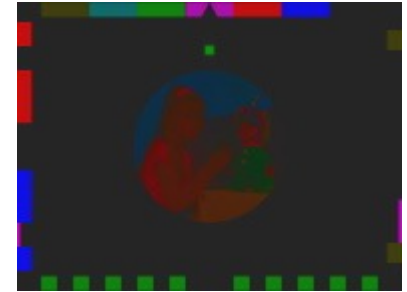
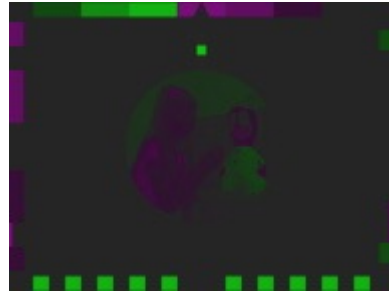


Luminance signal Y



Colour Difference signals R-Y, G-Y, B-Y

R, G, B and Colour Difference



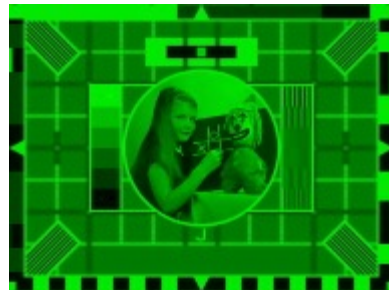
Colour difference displays – $(R-Y)$; $(G-Y)$; $(B-Y)$;
These show absence of the Y signal

$(R-Y)$, $(G-Y)$, $(B-Y)$

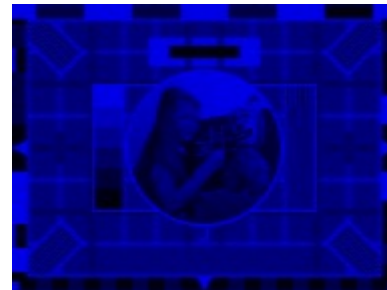
When Y is added, displays are:



R



G

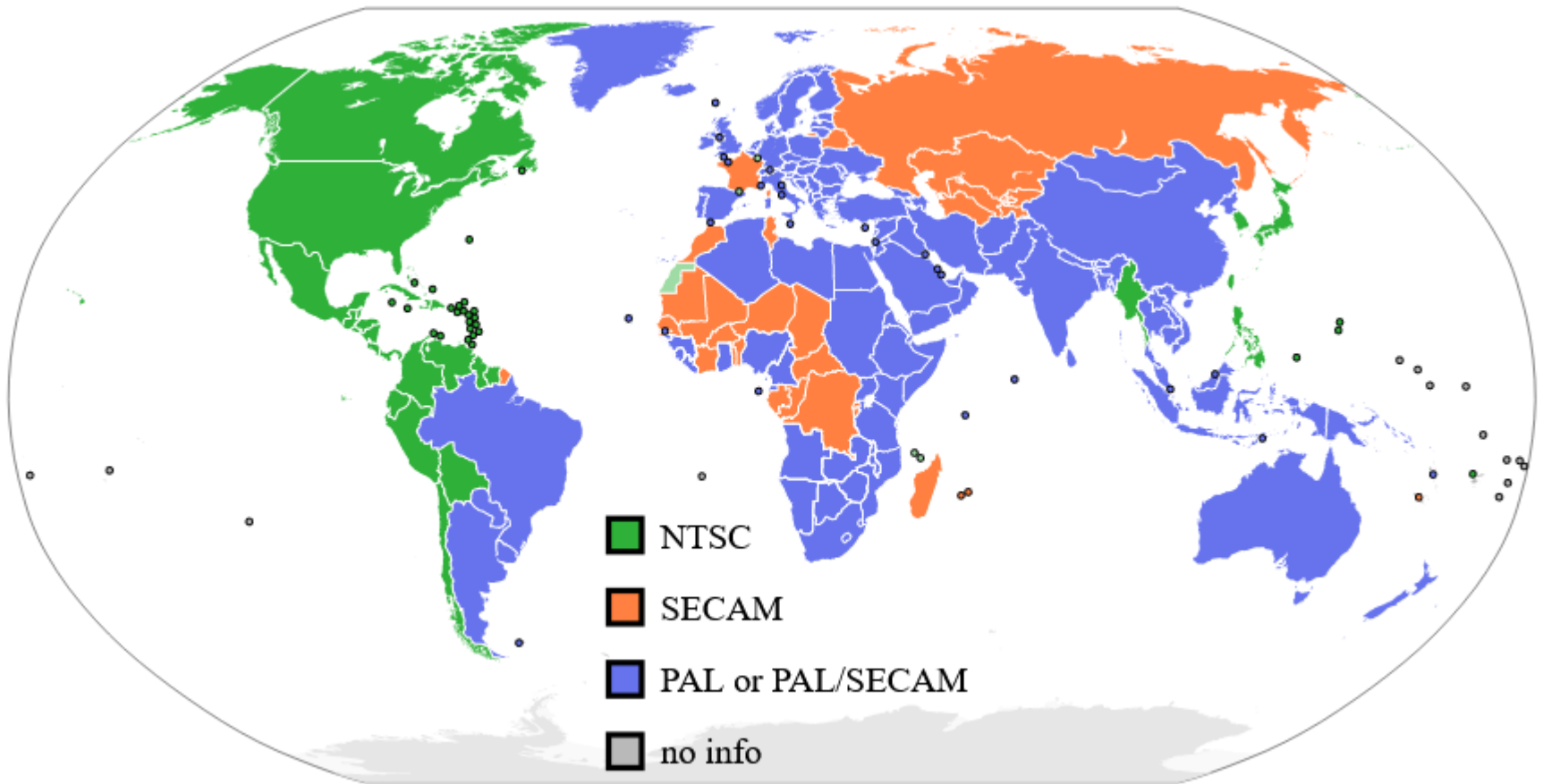


B



RGB

PAL and NTSC (and SECAM) in the Global Market



More on PAL and NTSC later in the module ...



B B C HD

1080 lines